# Four Principles of Corruption: A Systems-Theoretic Analysis of Institutional Failure and Reform

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**Abstract**

Public corruption is conventionally treated as a consequence of individual moral failure, addressed primarily through punitive enforcement. This framing has produced decades of anticorruption policy with limited systemic success. We propose an alternative: corruption is an emergent property of institutional systems operating under structural perturbation, not a deviation from an otherwise functional order but an adaptive response to dysfunction within that order. Drawing on evolutionary game theory, transaction-cost economics, information theory, and the theory of complex adaptive systems, we formulate four formal principles. The *Principle of Systemic Origin* establishes that corruption arises from institutional friction—the gap between formal rules and the conditions required for agents to achieve legitimate objectives—rather than from deficiencies in individual character. The *Principle of Systemic Coercion* demonstrates that institutional environments can render corrupt behavior the only individually rational strategy, making moral condemnation of participants analytically vacuous. The *Principle of Adaptive Morphology* proves that increasing punitive pressure does not reduce corruption but reshapes it into more complex, concealed, and costly forms, because enforcement alters the strategy space without eliminating the underlying perturbation. The *Principle of Elimination Through Automation* shows that replacing human decision-makers with algorithmic systems at corruption-susceptible nodes removes the necessary condition for corrupt transactions—namely, an agent with personal interests—thereby eliminating the phenomenon at its structural root. We further demonstrate, using results from the theory of totalizing ideals in complex adaptive systems, that the complete eradication of all informal adaptive behavior is not merely impractical but structurally damaging: it destroys the adaptive variance required for institutional viability under irreducible uncertainty. Each principle is supported by formal proofs, computational toy models with simulation results, and explicit consideration of counterarguments. The framework yields a diagnostic methodology: rather than asking *who is corrupt*, the productive question is *what structural perturbation makes corruption the rational adaptive response*, and *how can that perturbation be removed or its effects automated away.*

**Keywords:** public corruption, institutional failure, systems theory, game theory, transaction costs, adaptive systems, algorithmic governance, anticorruption reform

# Contents

# 1 Introduction

The dominant paradigm in anticorruption research and policy treats corruption as a principal-agent problem: a trusted official deviates from prescribed behavior to extract private rents, and the remedy consists of improved monitoring, harsher penalties, and stronger institutional oversight (Rose-Ackerman, 1999; Klitgaard, 1988; Becker, 1968). This framework has informed anticorruption programs worldwide for over four decades. Its empirical record is, at best, mixed. Cross-national studies consistently find that punitive enforcement campaigns produce short-term reductions in detected corruption followed by adaptation and resurgence (Persson et al., 2013; Mungiu-Pippidi, 2006). The question is why.

This paper argues that the limited success of conventional anticorruption approaches stems from a misdiagnosis. Corruption is not fundamentally a problem of deviant individuals operating within a sound institutional framework. It is, instead, an emergent property of institutional systems under structural stress—a predictable adaptive response by rational agents to the gap between formal rules and the conditions those rules create for the agents who must operate within them.

This reframing has a precise analog in the theory of self-organizing systems. In multi-agent systems where participants share an environment, interact repeatedly, and face finite resource constraints, cooperative behavior emerges as the statistically dominant and dynamically stable configuration (Axelrod, 1984; Nowak, 2006; Ostrom, 1990). Antagonistic or exploitative behavior—including corruption—does not arise spontaneously under such conditions. It requires and reflects the presence of an external perturbation: a structural feature of the institutional environment that distorts the payoff landscape, rendering cooperative strategies individually costly and defection individually rational (Kriger, 2026).

The concept of institutional friction provides the bridge between abstract systems theory and the concrete phenomenon of corruption. Transaction-cost economics (Coase, 1937; Williamson, 1979, 1985) has long recognized that economic activity is shaped not only by production costs but by the costs of transacting within institutional structures—the costs of obtaining permits, navigating regulations, securing approvals, and complying with procedural requirements. When these friction costs become sufficiently large, agents face a choice: absorb the costs (reducing their viability), exit the system (forgoing their objectives), or circumvent the friction through informal channels. Corruption, in this light, is the third option. It is the informal lubricant that agents apply to an institutional mechanism whose formal gears have become too stiff, too slow, or too misaligned to serve their intended function.

This observation does not constitute a defense of corruption. To identify the structural origins of a phenomenon is not to endorse it. Corruption carries severe costs: it distorts resource allocation, erodes institutional trust, entrenches inequality, and ultimately degrades the very system it temporarily circumvents (Mauro, 1995; Shleifer and Vishny, 1993; Bardhan, 1997). The point, rather, is diagnostic. If corruption arises from structural perturbation, then the effective remedy is not increased punishment of the agents who adapt to that perturbation, but identification and removal of the perturbation itself.

The paper develops this argument through four formal principles, each grounded in mathematical reasoning and supported by computational models.

The **Principle of Systemic Origin** (Section 3) establishes that corruption correlates with and is causally driven by institutional friction—the measurable gap between the formal requirements imposed on agents and the conditions under which those agents can achieve legitimate objectives. The principle is formalized through the concept of a perturbation operator that transforms a cooperative baseline payoff structure into one that incentivizes defection.

The **Principle of Systemic Coercion** (Section 4) demonstrates that under sufficiently strong perturbation, participation in corruption becomes the unique individually rational strategy. Agents who refuse to participate are systematically eliminated from the system through competitive disadvantage, not through any deliberate conspiracy, but through the ordinary operation of selection dynamics within the perturbed environment.

The **Principle of Adaptive Morphology** (Section 5) proves that increasing the severity of punishment for detected corruption does not reduce the total volume of corruption but alters its form. Agents adapt their strategies to the enforcement landscape, shifting from crude and detectable methods to sophisticated and concealed ones. The total social cost of corruption may increase under harsher enforcement regimes because the costs of concealment and the costs of enforcement both rise, while the underlying perturbation remains unaddressed.

The **Principle of Elimination Through Automation** (Section 6) identifies the structural precondition for any corrupt transaction: the presence of a human decision-maker with personal interests at the point where discretionary authority is exercised. Replacing this human agent with an algorithmic system removes the necessary condition for corruption. An automated system has no relatives to favor, no bribes to accept, no career to protect. The principle formalizes this observation and addresses the immediate counterargument—that algorithmic systems can themselves be corrupted at the design stage—by distinguishing between corruption at the point of execution and corruption at the point of design, and showing that the latter is a qualitatively different and more tractable problem.

Following the four principles, Section 7 introduces a structural constraint on anticorruption ambition. Drawing on the formally proved Asymmetry of Totalizing Ideals (Kriger, 2025), we demonstrate that the complete elimination of all informal adaptive behavior—including low-level corruption—constitutes a totalizing optimization that destroys the adaptive variance required for institutional viability under irreducible uncertainty. The pursuit of absolute institutional purity, like the pursuit of any terminal ideal in a complex adaptive system, generates greater long-term systemic damage than the maintenance of a bounded level of institutional flexibility. This result does not license corruption; it constrains the form that effective anticorruption reform can take.

Section 8 presents computational toy models that illustrate the dynamics described by each principle. These simulations are not offered as empirical evidence but as demonstrations that the formal mechanisms produce the predicted qualitative behavior under reasonable parameter choices.

Section 9 situates the framework within the existing literature on corruption, institutional economics, and governance reform, and addresses potential objections. Section 10 summarizes the results and outlines directions for future research.

### 1.1 Scope and Limitations

Several delimitations should be stated at the outset.

First, the formal framework developed here applies to institutional corruption—the systematic distortion of official decision-making processes through bribery, favoritism, extortion, and related practices. It does not directly address other forms of malfeasance such as fraud, embezzlement, or organized crime, though these may share structural features with institutional corruption.

Second, the mathematical models employ standard simplifying assumptions (rational agents, complete strategy spaces, well-defined payoff functions) that do not fully capture the psychological, cultural, and historical complexity of real-world corruption. The models are intended to identify structural tendencies, not to predict specific outcomes in specific contexts.

Third, the claim that corruption is systemically generated does not imply that individual agency is irrelevant. Agents make choices, and those choices have consequences. The argument is that the distribution of choices across a population is shaped primarily by the institutional structure within which choices are made, and that reform efforts targeting the structure will be more effective than those targeting individual decision-makers.

## 2 Formal Framework

This section establishes the mathematical apparatus used throughout the paper. The definitions are drawn from game theory, institutional economics, and the theory of complex adaptive systems, and are formulated to be accessible to readers from any of these fields.

### 2.1 Institutional System

An institutional system consists of agents operating within a structured environment defined by formal rules, resource constraints, and interaction patterns.

**Definition 2.1** (Institutional System)**.** An institutional system is a tuple $\mathcal{I} = (A, \Omega, \mathcal{R}, U)$ where:

- $A = \{a_1, a_2, \ldots, a_n\}$ is a finite set of agents (officials, citizens, firms);
- $\Omega$ is a state space describing the possible configurations of the system;
- $\mathcal{R} = \{r_1, r_2, \ldots, r_m\}$ is a set of formal rules governing agent behavior;
- $U : A \times A \to \mathbb{R}$ is a payoff function measuring the outcome of interactions between agents.

Each agent $a_i$ possesses a strategy set $S_i$ from which it selects actions. In an institutional context, strategies include both rule-compliant actions (following prescribed procedures, applying criteria as specified) and rule-circumventing actions (accepting bribes, granting favors, falsifying records). The payoff function $U$ captures the material and social consequences of strategy choices for interacting agents.

### 2.2 Institutional Friction

The central concept linking institutional structure to corrupt behavior is institutional friction: the measurable cost imposed on agents by the formal rules of the system, above and beyond what is structurally necessary for the system's stated objectives.

**Definition 2.2** (Institutional Friction)**.** Given an institutional system $\mathcal{I}$ with rule set $\mathcal{R}$, the institutional friction $\Phi(\mathcal{R})$ is:

$$\Phi(\mathcal{R}) = \sum_{j=1}^{m} c(r_j) - \sum_{j \in \mathcal{R}_E} c(r_j) \tag{1}$$

where $c(r_j)$ is the compliance cost of rule $r_j$, and $\mathcal{R}_E \subseteq \mathcal{R}$ is the subset of rules that are essential for the system's stated objectives (ensuring safety, preventing harm, maintaining fairness). The friction $\Phi$ measures the aggregate cost of discretionary rules—those whose removal would not compromise the system's core functions.

Institutional friction arises from multiple sources: redundant approval processes, unnecessary documentation requirements, artificially long waiting periods, opaque criteria for decisions, and bureaucratic procedures that serve no purpose beyond perpetuating the authority of the administering body. The essential/discretionary distinction is not always sharp in practice, but the conceptual separation is analytically necessary: some rules exist because the system's objectives require them, and others exist because the system's history, politics, or internal incentives have accumulated them. Friction, in this framework, refers specifically to the latter.

*Remark* 2.3 (Estimating $\Phi$ in Practice). The essential/discretionary distinction raises a practical measurement question: how would $\Phi(\mathcal{R})$ be estimated in a real institutional system? Several operational approaches are available. *Comparative benchmarking*: if two jurisdictions achieve the same policy objective (e.g., building safety) but one requires three approvals and the other requires twelve, the difference in compliance cost provides a lower bound on the friction in the more complex system. *Functional testing*: for each rule $r_j$, ask whether its removal would compromise the system's stated objective; rules that cannot be linked to any specific objective are candidates for the discretionary set $\mathcal{R} \setminus \mathcal{R}_E$. *User-reported burden*: systematic surveys of agents (citizens, firms, officials) about which procedures they find unnecessary or obstructive provide empirical data on perceived friction. None of these methods produces a precise value for $\Phi$, but they provide ordinal rankings across systems and over time, which is sufficient for the comparative-static predictions of the framework.

### 2.3 Perturbation Operator

When institutional friction is sufficiently large, it distorts the payoff landscape faced by agents. This distortion is formalized as a perturbation operator.

**Definition 2.4** (Perturbation Operator)**.** A perturbation operator $P : (A \times A \to \mathbb{R}) \to (A \times A \to \mathbb{R})$ modifies the baseline payoff structure $U_0$ to produce a perturbed payoff structure:

$$U_P(a_i, a_j) = U_0(a_i, a_j) + P(a_i, a_j) \tag{2}$$

The perturbation $P$ is *antagonistic* if it satisfies any of the following conditions:

(a) **Zero-sum injection:** $P(a_i, a_j) + P(a_j, a_i) \leq 0$ for all $i \neq j$;

(b) **Cooperation penalty:** $P(\text{comply}, \text{comply}) < 0$;

(c) **Defection subsidy:** $P(\text{circumvent}, \text{comply}) > 0$.

The perturbation operator captures the way institutional friction transforms the strategic environment. Consider a concrete instance: a building permit requires twelve separate approvals, each taking weeks. The baseline payoff for a developer who follows the rules includes both the value of the completed project and the costs of compliance. If those compliance costs are high enough, the perturbed payoff structure may satisfy condition (c): circumventing the process through a bribe yields a higher individual payoff than complying with it. The perturbation is not introduced by the corrupt agent; it is a feature of the institutional design.

Crucially, the perturbation $P$ must be structurally separable from the baseline game. It must be possible, at least in principle, to remove $P$—by simplifying rules, eliminating unnecessary approvals, reducing processing times—while keeping the underlying institutional system intact. This separability is what distinguishes structural perturbation from inherent features of the interaction. If a payoff asymmetry is intrinsic to the roles involved (as in employer–employee or regulator–regulated relationships), it is part of $U_0$, not $P$.

*Remark* 2.5 (Classification Uncertainty)*.* In practice, the boundary between essential and discretionary rules is not sharp. Some rules that appear discretionary exist because simpler alternatives were found to be exploitable—regulatory complexity in financial markets, for instance, often accumulates precisely because each layer addresses a genuine vulnerability. We formalize this uncertainty as follows. Let $q_j \in [0, 1]$ denote the probability that rule $r_j$ is correctly classified as discretionary. Then the expected friction is:

$$\mathbb{E}[\Phi(\mathcal{R})] = \sum_{j \notin \mathcal{R}_E} q_j \cdot c(r_j) \tag{3}$$

and the variance in friction estimation is:

$$\text{Var}[\Phi(\mathcal{R})] = \sum_{j \notin \mathcal{R}_E} q_j(1 - q_j) \cdot c(r_j)^2$$

The downstream results (Theorems 3.2–3.4) hold in expectation over this classification uncertainty: the expected corruption rate is monotonically increasing in expected friction. However, the precision of policy recommendations based on friction reduction depends on $\mathrm{Var}[\Phi]$: when classification uncertainty is high, the confidence interval around predicted corruption reduction is correspondingly wide. This is an irreducible limitation of the framework that constrains its quantitative (though not qualitative) applicability.

### 2.4 Corruption as Strategy

Within the perturbed payoff structure, corruption is formalized as a specific class of strategies.

**Definition 2.6** (Corrupt Strategy)**.** A strategy $s_i \in S_i$ is *corrupt* if it satisfies two conditions simultaneously:

(i) It violates at least one formal rule $r_j \in \mathcal{R}$;

(ii) It yields a higher individual payoff than any rule-compliant strategy under the perturbed payoff structure: $U_P(s_i^{\text{corrupt}}, s_{-i}) > U_P(s_i^{\text{comply}}, s_{-i})$ for the relevant strategy profile $s_{-i}$ of other agents.

This definition captures the essential duality of corruption: it is simultaneously a rule violation and a rational optimization. Neither component alone suffices. Rule-following that happens to be individually optimal is not corruption; it is compliance. Rule-breaking that reduces the agent's payoff is not corruption; it is error or protest. Corruption requires both the violation and the individual advantage, and it is the perturbation $P$ that creates the conditions under which these two features coincide.

### 2.5 Adaptive Variance

The final element of the framework concerns the system's capacity for learning and self-correction. This concept, drawn from information theory, will be needed in Section 7 to establish the structural limits of anticorruption reform.

**Definition 2.7** (Adaptive Variance)**.** Let $\pi_t \in \mathcal{P}(\Omega)$ denote the distribution over system states at time $t$, and let $\theta^* \in \Theta$ denote the true (unknown) parameter governing the environment. The adaptive variance is:

$$V_t := I(\pi_t; \theta^*) = H(\theta^*) - H(\theta^*|\pi_t) \tag{4}$$

where $I$ denotes mutual information, $H$ denotes Shannon entropy, and $H(\theta^*|\pi_t)$ is the conditional entropy of the environmental parameter given the state distribution.

Adaptive variance measures the amount of information that the diversity of system states carries about the unknown structure of the environment. A system with high adaptive variance maintains a range of different configurations—different institutional arrangements, different behavioral patterns, different local solutions—that collectively provide information about which approaches work and which do not. A system with zero adaptive variance has collapsed to a single configuration and has no internal basis for detecting whether that configuration is appropriate for the actual environment.

In institutional terms, adaptive variance corresponds to the flexibility inherent in a system that tolerates local variation, permits experimentation, and does not enforce rigid uniformity across all contexts. This concept will prove central to understanding why the complete elimination of all informal behavior—including minor corruption—can be structurally harmful.

### 2.6 Notation Summary

For reference, the principal notation used throughout the paper is collected here:

Table 1: Principal notation.

| Symbol | Meaning |
|---|---|
| $\mathcal{I} = (A, \Omega, \mathcal{R}, U)$ | Institutional system |
| $A = \{a_1, \ldots, a_n\}$ | Set of agents |
| $S_i$ | Strategy set of agent $a_i$ |
| $\mathcal{R}, \mathcal{R}_E$ | Full rule set, essential rule subset |
| $\Phi(\mathcal{R})$ | Institutional friction |
| $U_0$ | Baseline (unperturbed) payoff function |
| $P$ | Perturbation operator |
| $U_P = U_0 + P$ | Perturbed payoff function |
| $\pi_t$ | State distribution at time $t$ |
| $V_t = I(\pi_t; \theta^*)$ | Adaptive variance |
| $H(\cdot)$ | Shannon entropy |
| $\mu_i$ | Moral cost of corruption for agent $a_i$ |
| $\kappa$ | Concealment complexity |

# 3 Principle I: Systemic Origin

**Principle 1** (Systemic Origin)**.** The prevalence of corruption in an institutional system is a monotonically increasing function of institutional friction. Corruption arises not from deficiencies in the moral character of individual agents, but from the structural gap between formal institutional requirements and the conditions under which agents can achieve legitimate objectives. In the absence of institutional friction, the cooperative (non-corrupt) equilibrium is the unique stable attractor of the system.

The principle asserts a causal direction: institutional friction generates corruption, not the reverse. An institution with high friction and honest agents will develop corruption; an institution with low friction and dishonest agents will suppress it. The character of the agents is not irrelevant, but it is secondary to the structure within which they operate.

## 3.1 Formal Statement and Proof

We establish the principle through two results: first, that unperturbed peer systems converge to cooperative equilibria; second, that the introduction of friction-induced perturbation shifts the equilibrium toward corrupt strategies in proportion to the magnitude of the perturbation.

**Assumption 3.1** (Interaction Structure)**.** The institutional system satisfies: (i) agents interact repeatedly with a nontrivial probability of re-encountering previous partners ($\delta_{\text{repeat}} > 0$); (ii) agents can condition their behavior on observed partner history (identifiability); (iii) the interaction network has bounded average degree $k$.

These conditions are satisfied in institutional contexts where officials interact with recurring populations (regulatory agencies with repeat applicants, courts with regular attorneys, procurement offices with established vendors) but are *not* satisfied in one-shot, anonymous interactions (a tourist paying a traffic fine, a citizen visiting a government office once). The theorem's scope is explicitly limited to the former class.

**Theorem 3.2** (Cooperative Default Under Zero Friction)**.** *Consider an institutional system $\mathcal{I}$ with $n$ agents satisfying Assumption 3.1, interacting under a shared resource constraint $E$. Let $U_0$ be the baseline payoff structure with $\Phi(\mathcal{R}) = 0$ (zero institutional friction). Then, under Assumptions 3.3 and 3.1, the cooperative strategy profile constitutes a Nash equilibrium of the stage game, and the basin of attraction of this equilibrium in the replicator dynamics is strictly larger than that of any non-cooperative equilibrium.*

*Proof.* The proof proceeds in three steps.

*Step 1: Payoff structure under zero friction.* When $\Phi(\mathcal{R}) = 0$, the perturbation operator satisfies $P = 0$ (Definition 2.4), so $U_P = U_0$. Under the baseline payoff structure, compliant behavior incurs no excess cost beyond what is structurally necessary for the system's objectives. The payoff to compliance is:

$$U_0(s_i^{\text{comply}}, s_{-i}^{\text{comply}}) = b - c_E$$

where $b$ is the benefit of successful interaction and $c_E$ is the cost of complying with essential rules. The payoff to defection against a compliant population is:

$$U_0(s_i^{\text{corrupt}}, s_{-i}^{\text{comply}}) = b + \delta - p_d \cdot F$$

where $\delta$ is the immediate gain from circumventing rules, $p_d$ is the detection probability, and $F$ is the penalty upon detection.

We introduce an explicit assumption:

**Assumption 3.3** (Friction-Dependent Circumvention Gain)**.** The gain from circumvention $\delta$ is a function of institutional friction satisfying $\delta(0) = \delta_0 \geq 0$ and $\delta'(\Phi) > 0$, where $\delta_0$ is bounded and small relative to $p_d \cdot F$.

A note on this assumption is warranted. Even under zero discretionary friction, agents might gain by circumventing essential rules—for example, jumping a queue that exists for legitimate reasons. This means $\delta_0$ need not be exactly zero. The assumption requires only that $\delta_0 < p_d \cdot F$: the gain from circumventing essential rules alone is less than the expected penalty, which is a design requirement for any functional rule set. If essential rules are worth having, the penalties for violating them must exceed the gains from violation; otherwise the rules would not be enforceable even in principle. Under this assumption, the payoff to defection satisfies $b + \delta_0 - p_d \cdot F < b - c_E$ when $\delta_0 < p_d \cdot F - c_E$, making defection dominated.

*Step 2: Nash equilibrium.* Since $U_0(s_i^{\text{comply}}, s_{-i}^{\text{comply}}) > U_0(s_i^{\text{corrupt}}, s_{-i}^{\text{comply}})$ when $\delta_0 < p_d \cdot F$ (Assumption 3.3), no agent can improve its payoff by unilaterally switching from compliance to corruption. The all-comply profile is therefore a Nash equilibrium of the stage game under $U_0$.

*Step 3: Basin of attraction in evolutionary dynamics.* Under the replicator dynamics, the frequency $x_C$ of compliant agents evolves according to:

$$\frac{dx_C}{dt} = x_C(f_C - \bar{f}) \tag{5}$$

where $f_C$ is the fitness (expected payoff) of the compliant strategy and $\bar{f}$ is the population mean fitness. In repeated interactions with partner identification, the compliant strategy benefits from reciprocity: compliant agents preferentially interact with other compliant agents, yielding sustained mutual benefit. This network reciprocity effect, established in the evolutionary game theory literature (Nowak, 2006; Santos and Pacheco, 2005), amplifies the payoff advantage of cooperation. The benefit-to-cost ratio $b/c_E$ determines the critical threshold: when $b/c_E > k$ (where $k$ depends on the interaction network structure), cooperation is evolutionarily stable (Nowak, 2006). In peer systems with shared resources, $b$ is generically large (benefits of institutional function accrue to all) and $c_E$ is bounded (essential compliance costs are finite and purposeful), so $b/c_E > k$ holds generically.

The basin of attraction of the cooperative equilibrium—the set of initial conditions from which the replicator dynamics converge to full cooperation—is therefore strictly larger than that of any defection equilibrium, which requires either one-shot interaction (ruling out reciprocity), anonymity (ruling out partner identification), or $b/c_E < k$ (incompatible with the shared-resource structure of institutional systems). □

The theorem establishes the baseline: in the absence of unnecessary friction, cooperation is the natural attractor. The next result shows what happens when friction is introduced.

**Theorem 3.4** (Friction-Corruption Monotonicity)**.** *Let $\gamma(\mathcal{I})$ denote the equilibrium proportion of agents employing corrupt strategies in institutional system $\mathcal{I}$. Then $\gamma$ is a monotonically increasing function of institutional friction:*

$$\frac{\partial \gamma}{\partial \Phi} > 0 \tag{6}$$

*for all $\Phi > 0$ and under reasonable regularity conditions on the payoff structure.*

*Proof.* Consider the perturbed payoff structure $U_P = U_0 + P$, where $P$ is generated by institutional friction $\Phi$. The gain from corruption is:

$$\Delta(\Phi) := U_P(s_i^{\text{corrupt}}, s_{-i}) - U_P(s_i^{\text{comply}}, s_{-i})$$

Expanding:

$$\Delta(\Phi) = \underbrace{[\delta(\Phi) - p_d \cdot F]}_{\text{direct gain net of risk}} + \underbrace{[c_D(\Phi) - c_E]}_{\text{avoided friction cost}} \tag{7}$$

where $\delta(\Phi)$ is the gain from circumvention (increasing in $\Phi$, since higher friction creates more valuable shortcuts) and $c_D(\Phi)$ is the compliance cost under discretionary rules (increasing in $\Phi$ by definition of friction).

The term $c_D(\Phi) - c_E$ is precisely the excess burden imposed by discretionary rules. Under the simplifying assumption that friction costs are distributed uniformly across agents, each agent bears $c_D(\Phi) - c_E = \Phi/n$. (This uniform distribution assumption can be relaxed: the monotonicity result holds for any distribution of friction costs across agents, provided that total friction $\Phi$ increases each agent's burden in expectation.) Both $\delta(\Phi)$ and $c_D(\Phi)$ are increasing in $\Phi$, so:

$$\frac{\partial \Delta}{\partial \Phi} > 0$$

In the replicator dynamics (5), the fitness of the corrupt strategy relative to compliance increases with $\Delta(\Phi)$. By standard results on the replicator equation, the equilibrium frequency of a strategy is a monotonically increasing function of its relative fitness (Nowak, 2006). Therefore:

$$\frac{\partial \gamma}{\partial \Phi} = \frac{\partial \gamma}{\partial \Delta} \cdot \frac{\partial \Delta}{\partial \Phi} > 0$$

since both partial derivatives are strictly positive. □

Together, Theorems 3.2 and 3.4 establish the first principle: corruption is zero at zero friction and increases monotonically with friction. The implication for policy is direct: reducing institutional friction is a structurally guaranteed method of reducing corruption, while leaving friction in place and increasing punishment addresses only the detection term $p_d \cdot F$ in equation (7), without affecting the structural incentive $\Delta(\Phi)$.

### 3.2 Illustrative Empirical Evidence

The formal results are consistent with—though not proven by—a substantial body of empirical evidence. We present the following observations as illustrative cases, not as systematic empirical tests. Each involves confounding variables that prevent clean causal attribution; their value lies in qualitative consistency with the theoretical predictions, and they suggest directions for future empirical investigation.

Cross-national studies have repeatedly found that corruption correlates more strongly with regulatory burden, bureaucratic complexity, and institutional opacity than with cultural variables or enforcement intensity (Treisman, 2000; Svensson, 2005; Dimant and Tosato, 2018). These correlations are consistent with Theorem 3.4 but do not establish causation, as regulatory burden and corruption may share common causes (weak state capacity, historical path dependence, political instability).

The transition economies of the 1990s provide a suggestive, though not conclusive, natural experiment. Countries that implemented rapid deregulation and simplification of business procedures (notably Estonia and Georgia) experienced significant reductions in corruption, while countries that maintained complex regulatory structures but increased enforcement (notably Russia and Ukraine in the same period) saw corruption shift in form without diminishing in volume (Olken and Pande, 2012). However, these transitions involved many simultaneous changes—EU accession processes, geopolitical realignment, differences in initial conditions and state capacity—that make it impossible to attribute outcomes to friction reduction alone. The observation is consistent with the framework but does not constitute a controlled test of it.

The introduction of electronic government services—which reduces friction by automating routine approvals and eliminating discretionary decision points—has been associated with measurable corruption reductions in countries as diverse as South Korea, India, and Estonia (Rothstein, 2011).

The historical trajectory of corruption also supports the principle. In pre-modern societies, where formal institutional structures were minimal, the relationships that we would now classify as corrupt—exchange of favors, patronage networks, gift-giving to officials—were the normal mode of social organization (North, 1990). Corruption as a distinct category emerged only with the development of formal rules, and its prevalence tracks the gap between the complexity of those rules and their alignment with agents' legitimate needs.

### 3.3 Counterarguments and Responses

#### *3.3.1 Objection: Low-Friction Systems Still Exhibit Corruption*

Some institutional systems with relatively low bureaucratic burden still experience corruption (e.g., campaign finance manipulation in established democracies, insider trading in efficient financial markets). Does this refute the principle?

*Response.* The principle asserts that corruption is a monotonically increasing function of friction, not that friction is the sole cause. Low-friction systems with residual corruption exhibit one or more of the following: (a) hidden friction in specific subsystems (e.g., campaign finance rules that are nominally simple but create complex compliance requirements); (b) friction of a non-bureaucratic kind (information asymmetries, access barriers, network-dependent decision-making); or (c) perturbation arising from sources other than rule complexity (e.g., extreme resource concentration creating power asymmetries that function as $P$). The principle accommodates these cases through the generality of the perturbation operator $P$, which captures any structural feature that distorts the payoff landscape away from the cooperative baseline.

#### *3.3.2 Objection: Some Cultures Are More Corrupt Regardless of Institutions*

A persistent claim in popular and some academic discourse holds that corruption is culturally determined—that some societies are "simply more corrupt" due to cultural norms, values, or traditions.

*Response.* Cultural norms around corruption are better understood as internalized perturbation—the residual effect of historical institutional friction that persists even after the original friction has been reduced. This is formally analogous to the concept of internalized perturbation $P_{\text{int}}$ in the theory of self-organizing systems (Kriger, 2026): when an external perturbation is maintained for a sufficient period, agents internalize the perturbed payoff structure through cultural transmission, and this internalization decays only slowly after the external perturbation is removed. The prediction is specific and testable: cultural norms favoring corruption will erode over generational timescales when institutional friction is reduced, even without direct cultural intervention. The experience of countries such as Botswana, Chile, and Estonia—which achieved significant corruption reduction through institutional reform without cultural revolution—supports this prediction.

#### *3.3.3 Objection: The Principle Absolves Individuals of Responsibility*

If corruption is systemically generated, are individual corrupt actors blameless?

*Response.* The principle is a diagnostic tool, not a moral exculpation. It identifies the structural conditions that generate corruption at the population level. Individual agents retain the capacity for choice and may be held accountable for their choices within the existing legal framework. The point is not that individuals are blameless but that policies focused exclusively on individual accountability will fail to reduce aggregate corruption if the structural conditions generating it remain in place. A physician who identifies a pathogen as the cause of an epidemic does not thereby absolve individual patients of responsibility for hygiene; the physician merely points out that hygiene campaigns alone will not end the epidemic if the pathogen's reservoir is not addressed.

# 4 Principle II: Systemic Coercion

**Principle 2** (Systemic Coercion)**.** When institutional friction exceeds a critical threshold, participation in corruption becomes the unique individually rational strategy for agents occupying discretionary positions. Agents who refuse to participate are systematically displaced through competitive selection. Moral condemnation of individual participants is therefore analytically insufficient as a basis for reform, since the observed behavior reflects the structure of the system rather than the character of its occupants.

Where the first principle establishes that institutional friction generates corruption, the second principle identifies the mechanism by which a corrupt equilibrium, once established, becomes self-reinforcing. The system does not merely permit corruption; at sufficient friction levels, it compels it.

## 4.1 Formal Statement and Proof

**Theorem 4.1** (Critical Friction Threshold)**.** *There exists a critical friction level $\Phi^*$ such that for all $\Phi > \Phi^*$, the corrupt strategy is the unique best response for any agent occupying a discretionary position, regardless of the agent's initial moral preferences.*

To prove this theorem, we first formalize the role of moral preferences within the payoff structure.

**Definition 4.2** (Moral Cost)**.** Each agent $a_i$ possesses a moral cost parameter $\mu_i \geq 0$ representing the psychological disutility of engaging in corrupt behavior. The effective payoff of corruption for agent $a_i$ is:

$$U_P^{\text{eff}}(s_i^{\text{corrupt}}, s_{-i}) = U_P(s_i^{\text{corrupt}}, s_{-i}) - \mu_i \tag{8}$$

The population distribution of moral costs is $\mu \sim G(\cdot)$ with support on $[0, \bar{\mu}]$, where $\bar{\mu} < \infty$ is the maximum moral cost in the population.

The moral cost parameter captures the fact that individuals differ in their psychological resistance to corruption. Some agents experience significant discomfort when violating norms; others experience little. The critical point is that $\bar{\mu}$ is finite: no agent has infinite moral resistance. This is an empirical claim about human psychology, not a cynical assumption—it merely states that every person has some threshold beyond which the costs of compliance exceed the moral cost of circumvention.

**Assumption 4.3** (Bounded Exit)**.** Agents face an exit cost $c_{\text{exit}} > 0$ for leaving the institutional system (forgoing career, relocation costs, opportunity costs of alternative employment). An agent exits rather than participates in corruption when $\mu_i > \Delta(\Phi)$ and $c_{\text{exit}} < \mu_i - \Delta(\Phi)$. We assume that for the population of agents who have self-selected into discretionary positions, the effective maximum moral resistance (accounting for exit) satisfies $\bar{\mu}_{\text{eff}} < \infty$.

This assumption deserves scrutiny. In practice, agents can exit the system rather than participate in corruption, which bounds the effective friction they face. The proof therefore applies to the population of agents who remain within the system—those for whom exit costs exceed the gap between moral resistance and the structural advantage of corruption. The higher the exit costs (specialized career investment, geographic constraints, limited alternative employment), the larger this population and the more strongly the theorem applies. In institutional environments where exit is easy and alternatives abundant, the theorem's force is weakened; in environments where agents are locked in, it applies with full strength.

*Proof of Theorem 4.1.* The proof identifies the friction level at which even the most morally resistant agent finds corruption individually rational.

*Step 1: Net advantage of corruption.* From equation (7), the gain from corruption for agent $a_i$ net of moral cost is:

$$\Delta_i(\Phi) = \Delta(\Phi) - \mu_i = [\delta(\Phi) - p_d \cdot F] + [c_D(\Phi) - c_E] - \mu_i \tag{9}$$

*Step 2: Threshold for the maximally resistant agent.* The agent with the highest moral cost is $a^* = \arg\max_i \mu_i$, with $\mu_{a^*} = \bar{\mu}$. This agent switches to corruption when $\Delta_{a^*}(\Phi) > 0$, which requires:

$$\Phi^* : \quad \Delta(\Phi^*) = \bar{\mu} \tag{10}$$

Since $\Delta(\Phi)$ is continuous and monotonically increasing in $\Phi$ (Theorem 3.4), and $\Delta(0) < 0$ (corruption is disadvantageous under zero friction), and $\lim_{\Phi\to\infty} \Delta(\Phi) = \infty$ (friction costs grow without bound), the intermediate value theorem guarantees the existence and uniqueness of $\Phi^*$.

*Step 3: Universality above threshold.* For all $\Phi > \Phi^*$ and for all agents $a_i$ with $\mu_i \leq \bar{\mu}$:

$$\Delta_i(\Phi) = \Delta(\Phi) - \mu_i \geq \Delta(\Phi) - \bar{\mu} > \Delta(\Phi^*) - \bar{\mu} = 0$$

Therefore corruption is the unique best response for every agent in the population. □

The theorem establishes the existence of a tipping point. Below $\Phi^*$, some agents comply and others defect, depending on their individual moral costs. Above $\Phi^*$, the structural incentive overwhelms even the strongest moral resistance. The population composition becomes irrelevant: replacing corrupt agents with honest ones will not change the equilibrium outcome, because the honest replacements face the same payoff structure and will eventually adopt the same strategy.

**Theorem 4.4** (Selection Dynamics and Displacement). *In an institutional system with $\Phi > \Phi^*$ and competitive selection among agents occupying discretionary positions, non-corrupt agents are displaced at rate proportional to their competitive disadvantage. The expected time to full displacement of compliant agents is finite.*

*Proof.* Consider two agents, $a_C$ (compliant) and $a_D$ (corrupt), competing for the same discretionary position. Their accumulated resources after $T$ periods are:

$$W_C(T) = \sum_{t=1}^{T} U_P(s^{\text{comply}}, s_{-i}^{(t)}) \tag{11}$$

$$W_D(T) = \sum_{t=1}^{T} \left[U_P(s^{\text{corrupt}}, s_{-i}^{(t)}) - \mu_D\right] \tag{12}$$

Since $\Phi > \Phi^*$, by the proof of Theorem 4.1, $U_P(s^{\text{corrupt}}, s_{-i}) - \mu_D > U_P(s^{\text{comply}}, s_{-i})$ for every period and every $\mu_D \leq \bar{\mu}$. The wealth gap grows linearly:

$$W_D(T) - W_C(T) = \sum_{t=1}^{T} [\Delta(\Phi) - \mu_D] = T \cdot [\Delta(\Phi) - \mu_D] \tag{13}$$

In institutional contexts, accumulated resources translate into competitive advantage through multiple channels: political influence, network building, resource allocation, and patronage. Let $p_{\text{retain}}(W)$ denote the probability that an agent retains their position as a function of accumulated wealth. Under any reasonable selection mechanism where $p_{\text{retain}}$ is increasing in $W$, the compliant agent faces a steadily growing probability of displacement:

$$\mathbb{P}(a_C \text{ displaced by time } T) = 1 - \prod_{t=1}^{T} \frac{p_{\text{retain}}(W_C(t))}{p_{\text{retain}}(W_D(t))} \to 1 \quad \text{as } T \to \infty \tag{14}$$

The convergence to 1 is guaranteed by the linear growth of the wealth gap (13), which ensures that the ratio $p_{\text{retain}}(W_C)/p_{\text{retain}}(W_D)$ is bounded below 1 for all sufficiently large $t$. ∎

The displacement mechanism operates without conspiracy or deliberate exclusion. It is a consequence of competitive dynamics in a perturbed environment. Compliant agents accumulate fewer resources, build weaker networks, and exercise less influence than their corrupt counterparts. Over time, they are outcompeted, marginalized, and eventually replaced—not because the system targets them, but because the system's incentive structure systematically favors their competitors.

This is why moral condemnation of corrupt officials is analytically insufficient. The individuals who occupy corrupt positions are the survivors of a selection process that filters for willingness to participate in corruption. Replacing them with individuals of higher moral character merely restarts the selection process; the replacements will either adapt to the system's demands or be displaced in turn.

*Remark* 4.5 (Estimating $\Phi^*$ in Practice). The theorem's policy relevance depends on whether real institutional systems exceed $\Phi^*$. Estimating $\Phi^*$ requires knowledge of two quantities: the gain function $\Delta(\Phi)$ and the maximum moral resistance $\bar{\mu}$ in the population. Neither is directly observable, but indirect estimation is feasible.

For $\Delta(\Phi)$: the structural advantage of corruption can be estimated from the price of bribes (which reflects the value of circumventing friction to the payer) and from time-cost comparisons between formal and informal channels for the same service.

For $\bar{\mu}$: experimental economics provides methods for eliciting willingness to engage in rule-violation under controlled payoff structures (Fehr and Gächter, 2002). The distribution $G(\cdot)$ can be estimated from such experiments, though cross-cultural variation is substantial.

The threshold $\Phi^*$ can also be estimated indirectly from observed corruption prevalence data. If the framework is correct, jurisdictions with corruption rates approaching 100% among officials in specific agencies are likely above $\Phi^*$ for those agencies, while jurisdictions with low corruption rates among comparable officials are likely below $\Phi^*$. This approach is circular if used to validate the framework, but useful for calibration once the framework is accepted.

We acknowledge that in the absence of precise estimates of $\Phi^*$, the theorem's practical force is primarily qualitative: it establishes the existence and mechanism of a threshold phenomenon, not its quantitative location in specific institutional contexts. Quantitative calibration is a priority for future empirical work.

### 4.2 Illustrative Empirical Evidence

The displacement mechanism is well-documented in studies of police corruption, where officers who refuse to participate in systemic corruption are ostracized, denied promotions, and eventually forced out (Rose-Ackerman, 1999). Similar dynamics have been observed in public procurement, where firms that refuse to pay bribes are systematically excluded from contract competitions, and in regulatory agencies, where officials who insist on strict rule enforcement are reassigned to marginal positions.

The anticorruption literature has increasingly recognized that corruption functions as a collective action problem rather than an individual choice problem (Persson et al., 2013). The contribution of the present framework is to formalize this insight: the collective action failure is not a spontaneous coordination problem but a predictable consequence of institutional friction exceeding the critical threshold $\Phi^*$.

### 4.3 Counterarguments and Responses

#### *4.3.1 Objection: Some Individuals Resist Corruption Even in Highly Corrupt Systems*

Individual whistleblowers and reformers do exist in highly corrupt environments, apparently contradicting the claim that corruption becomes the unique rational strategy.

*Response.* The existence of individual resisters does not contradict the principle; it illustrates its selection dynamics. Whistleblowers and reformers in highly corrupt systems typically face severe consequences: career destruction, social isolation, physical danger, and in extreme cases, assassination. Their existence is the exception that confirms the rule: the system is structured to eliminate them, and the vast majority are in fact eliminated. Those who survive do so through external protection (international organizations, media attention, witness protection programs), which constitutes an exogenous intervention that partially offsets the perturbation $P$. The principle does not assert that resistance is impossible; it asserts that resistance is individually costly, that the cost increases with $\Phi$, and that selection dynamics ensure resisters are progressively removed from the population of institutional actors.

#### *4.3.2 Objection: The Principle Implies Fatalism About Corruption*

If the system compels corruption, is reform futile?

*Response.* The principle implies the opposite of fatalism: it identifies the precise lever for reform. If corruption is compelled by institutional friction, then reducing friction—simplifying procedures, automating routine decisions, increasing transparency, aligning rules with legitimate objectives—will reduce corruption by lowering $\Phi$ below $\Phi^*$. The fatalistic conclusion follows only from the assumption that institutional structures are immutable, which is empirically false. The principle does imply that reforms targeting individual behavior (ethics training, integrity pledges, exemplary punishments) will fail unless accompanied by structural change, because such reforms attempt to alter agent preferences ($\mu_i$) without altering the payoff structure ($\Delta(\Phi)$) that overwhelms those preferences.

#### *4.3.3 Objection: Moral Norms Can Shift the Threshold*

If society cultivates stronger moral norms (increasing $\bar{\mu}$), does this not raise $\Phi^*$ and make the system more resistant to corruption?

*Response.* In principle, yes: increasing $\bar{\mu}$ raises $\Phi^*$ by equation (10). In practice, however, the scope for increasing $\bar{\mu}$ is limited by human psychology, while the scope for reducing $\Phi$ through institutional design is substantial. Furthermore, the attempt to increase $\bar{\mu}$ through moral exhortation is itself subject to diminishing returns: moral campaigns in environments with high structural friction generate cynicism rather than compliance, because agents correctly perceive a contradiction between the moral demand and the structural incentive. The more effective strategy is to reduce friction to the point where compliance is individually rational without requiring exceptional moral fortitude—a design objective rather than a moral aspiration.

## 5 Principle III: Adaptive Morphology

**Principle 3** (Adaptive Morphology)**.** Increasing the severity of punishment for detected corruption does not reduce the total volume of corrupt transactions. It transforms the strategy space, driving corruption from crude and detectable forms into sophisticated, concealed, and costly forms. The total social cost of corruption—comprising the direct costs of corrupt transactions, the costs of concealment, and the costs of enforcement—may increase under harsher punishment regimes when the underlying institutional friction remains unaddressed.

This principle challenges the foundational assumption of deterrence-based anticorruption policy: that sufficiently severe punishment will reduce the incidence of corruption. The formal argument shows that deterrence operates on the detection term of the payoff equation while leaving the structural incentive intact. Rational agents respond not by ceasing corrupt behavior but by investing in concealment, effectively shifting the form of corruption without reducing its substance.

## 5.1 Formal Statement and Proof

We model corruption strategies as varying along a complexity dimension that determines both their concealment effectiveness and their cost.

**Definition 5.1** (Strategy Complexity)**.** Each corrupt strategy $s_i^{\text{corrupt}}$ is characterized by a complexity parameter $\kappa \in [0, \bar{\kappa}]$, where $\bar{\kappa}$ may be finite or infinite depending on the institutional context, and where:

- $p_d(\kappa)$ is the detection probability, a strictly decreasing function of $\kappa$: $p_d'(\kappa) < 0$, with $p_d(0) = p_{\max} \leq 1$ and $\lim_{\kappa \to \bar{\kappa}} p_d(\kappa) = \underline{p}_d \geq 0$;
- $c_\kappa(\kappa)$ is the concealment cost, a strictly increasing function of $\kappa$: $c_\kappa'(\kappa) > 0$, with $c_\kappa(0) = 0$.

When $\bar{\kappa} = \infty$ and $\underline{p}_d = 0$, the concealment technology is "rich" in the sense of Remark 5.4. When $\bar{\kappa} < \infty$ or $\underline{p}_d > 0$, the concealment technology has a ceiling, and the volume invariance result of Theorem 5.3(ii) holds only partially (see the Limitations discussion in Section 9).

Strategy complexity captures the sophistication of corruption methods. A bribe handed across a desk in cash has $\kappa \approx 0$: easy to detect, zero concealment cost. A network of shell companies routing payments through multiple jurisdictions has high $\kappa$: difficult to detect, expensive to maintain. The definitions above state that more complex strategies are harder to detect but costlier to execute, which accords with empirical observation.

**Definition 5.2** (Enforcement Intensity)**.** The enforcement regime is characterized by a penalty parameter $F > 0$. The expected cost of detection for a corrupt agent using strategy complexity $\kappa$ is:

$$C_{\text{det}}(\kappa, F) = p_d(\kappa) \cdot F \tag{15}$$

**Theorem 5.3** (Punishment Induces Complexity, Not Reduction)**.** *Let $\kappa^*(F)$ denote the optimal strategy complexity chosen by a rational corrupt agent facing penalty $F$. Under the assumption that the concealment technology is sufficiently rich (i.e., $\lim_{\kappa \to \infty} p_d(\kappa) = 0$—detection probability can be driven arbitrarily close to zero through sufficient investment):*

*(i)* *$\kappa^*(F)$ is strictly increasing in $F$: $\frac{d\kappa^*}{dF} > 0$;*

*(ii)* *The total volume of corrupt transactions $\gamma(\Phi, F)$ is approximately invariant to $F$ when $\Phi$ is held constant: $\frac{\partial \gamma}{\partial F}\big|_{\Phi = const} \approx 0$;*

*(iii)* *The total social cost of corruption $\Sigma(F)$ is increasing in $F$ when $\Phi > \Phi^*$: $\frac{d\Sigma}{dF} > 0$.*

*Remark* 5.4 (Scope Condition on Concealment Technology). The volume invariance result in part (ii) depends critically on the assumption that detection probability can be driven to zero through concealment investment. This assumption holds in institutional environments with complex financial systems, multiple jurisdictions, and rich intermediation networks—precisely the environments where sophisticated corruption is most problematic. In institutional contexts where concealment options are structurally limited—for example, small communities with high mutual visibility, or highly standardized transactions with automated monitoring—detection probability may have a hard floor $\underline{p}_d > 0$ that no amount of concealment can overcome. In such contexts, sufficiently severe punishment can reduce corruption volume, and part (ii) does not apply. The theorem is therefore strongest in the environments where anticorruption policy is most needed: large-scale, complex institutional systems with rich strategy spaces for concealment.

*Proof. Part (i): Optimal complexity increases with penalty.*

A rational agent choosing to engage in corruption selects the complexity level $\kappa$ that maximizes the net payoff:

$$\kappa^*(F) = \arg\max_{\kappa \geq 0} [\Delta(\Phi) - c_\kappa(\kappa) - p_d(\kappa) \cdot F] \tag{16}$$

where $\Delta(\Phi) > 0$ is the structural advantage of corruption (from Theorem 3.4).

The first-order condition is:

$$-c'_\kappa(\kappa^*) - p'_d(\kappa^*) \cdot F = 0 \tag{17}$$

Rearranging:

$$c'_\kappa(\kappa^*) = -p'_d(\kappa^*) \cdot F = |p'_d(\kappa^*)| \cdot F \tag{18}$$

This equation states that the agent equates the marginal cost of additional concealment with the marginal reduction in expected penalty. The second-order condition is:

$$-c''_\kappa(\kappa^*) - p''_d(\kappa^*) \cdot F < 0$$

which holds when $c_\kappa$ is convex and $p_d$ is concave or linear in $\kappa$ (standard assumptions for cost and detection functions).

To find $d\kappa^*/dF$, apply the implicit function theorem to (17):

$$\frac{d\kappa^*}{dF} = \frac{|p'_d(\kappa^*)|}{c''_\kappa(\kappa^*) + |p''_d(\kappa^*)| \cdot F} > 0 \tag{19}$$

The numerator is positive (detection probability decreases with complexity) and the denominator is positive (by the second-order condition). Therefore $\kappa^*$ is strictly increasing in $F$: harsher penalties induce more sophisticated concealment.

*Part (ii): Approximate volume invariance.*

The decision to engage in corruption at all depends on whether the net payoff at optimal complexity is positive:

$$\Pi^*(\Phi, F) := \Delta(\Phi) - c_\kappa(\kappa^*(F)) - p_d(\kappa^*(F)) \cdot F > 0 \tag{20}$$

By the envelope theorem, the effect of $F$ on the optimized payoff is:

$$\frac{\partial \Pi^*}{\partial F} = -p_d(\kappa^*(F)) \tag{21}$$

As $F$ increases, $\kappa^*$ increases, driving $p_d(\kappa^*(F))$ toward zero. The rate of decrease of $\Pi^*$ with respect to $F$ therefore diminishes:

$$\frac{\partial^2 \Pi^*}{\partial F^2} = -p_d'(\kappa^*) \cdot \frac{d\kappa^*}{dF} > 0 \tag{22}$$

This convexity implies that $\Pi^*$ decreases at a diminishing rate as $F$ increases. For $\Phi$ sufficiently above $\Phi^*$, $\Delta(\Phi)$ is large enough that $\Pi^*$ remains positive for all finite $F$: the structural gain from corruption exceeds the sum of concealment costs and residual expected penalties, regardless of penalty severity. Consequently, the proportion of agents choosing corruption $\gamma$ depends on $\Phi$ (which determines how many agents find corruption advantageous) but not on $F$ (which only affects the form of their corrupt strategies).

More precisely, the participation threshold is the moral cost $\mu^*$ at which $\Pi^*(\Phi, F) = \mu^*$. The change in corruption volume as $F$ increases is:

$$\frac{\partial \gamma}{\partial F} = g(\mu^*) \cdot \left| \frac{\partial \Pi^*}{\partial F} \right| = g(\mu^*) \cdot p_d(\kappa^*(F)) \tag{23}$$

where $g(\cdot)$ is the density of the moral cost distribution $G$ evaluated at the marginal participant.

This yields an explicit bound on the sensitivity of corruption volume to punishment:

$$\left| \frac{\partial \gamma}{\partial F} \right| \leq g_{\max} \cdot p_d(\kappa^*(F)) \tag{24}$$

where $g_{\max} = \sup_\mu g(\mu)$ is the maximum density of the moral cost distribution.

As $F$ increases and $\kappa^*$ adapts, $p_d(\kappa^*(F)) \to 0$ under the rich concealment technology assumption (Remark 5.4), so $\partial\gamma/\partial F \to 0$. The volume response vanishes asymptotically but may be non-negligible for moderate $F$ if $g_{\max}$ is large—that is, if many agents have moral costs clustered near the participation threshold.

Equation (24) makes the claim precise: volume invariance holds tightly when the concealment technology is rich (small $p_d$ at equilibrium) or when the moral cost distribution is diffuse (small $g_{\max}$). It holds loosely when concealment options are limited or when moral costs are concentrated near the threshold. The claim is therefore not a universal law but a structural tendency whose quantitative force depends on measurable properties of the institutional environment and agent population.

*Part (iii): Total social cost increases.*

The total social cost of corruption comprises three components:

$$\Sigma(F) = \underbrace{\gamma \cdot L_{\text{direct}}}_{\text{allocative distortion}} + \underbrace{\gamma \cdot c_\kappa(\kappa^*(F))}_{\text{concealment waste}} + \underbrace{C_{\text{enforce}}(F)}_{\text{enforcement expenditure}} \tag{25}$$

where $L_{\text{direct}}$ is the per-transaction allocative loss from corrupt decisions, $\gamma \cdot c_\kappa(\kappa^*)$ is the aggregate resource expenditure on concealment, and $C_{\text{enforce}}(F)$ is the cost of maintaining the enforcement apparatus (investigation, prosecution, adjudication).

Since $\gamma$ is approximately constant in $F$, $\kappa^*$ is increasing in $F$, $c_\kappa$ is increasing in $\kappa$, and $C_{\text{enforce}}$ is increasing in $F$ (detecting more sophisticated corruption requires more sophisticated enforcement), all three components of $\Sigma$ are non-decreasing in $F$, with at least two strictly increasing:

$$\frac{d\Sigma}{dF} = \gamma \cdot c'_\kappa(\kappa^*) \cdot \frac{d\kappa^*}{dF} + \frac{dC_{\text{enforce}}}{dF} > 0$$

□

The theorem formalizes a cycle that is well recognized by practitioners but rarely modeled: enforcement escalation produces concealment escalation, which produces further enforcement escalation, with each round increasing the total cost to society while leaving the underlying volume of corruption essentially unchanged.

### 5.2 Corollary: The Arms Race Dynamics

**Corollary 5.5** (Enforcement–Concealment Arms Race)**.** *In a dynamic setting where enforcement intensity $F_t$ and strategy complexity $\kappa_t$ co-evolve, the system converges to* *a steady state characterized by:*

$$\lim_{t\to\infty}[c_\kappa(\kappa_t) + C_{enforce}(F_t)] = \Lambda(\Phi) \tag{26}$$

*where* $\Lambda(\Phi)$ *is an increasing function of institutional friction. The waste from the arms race is bounded below by* $\Lambda(\Phi) - \Lambda(0)$*, representing pure deadweight loss attributable to unaddressed friction.*

This corollary identifies the concealment–enforcement arms race as a pure deadweight loss: resources expended on concealing and detecting corruption produce no social value. The magnitude of this waste is determined by $\Phi$, not by $F$. Increasing enforcement does not reduce the waste; it redistributes it between concealment and enforcement categories while increasing the total.

### 5.3 Empirical Support

The adaptive morphology of corruption is extensively documented. The transition from direct cash bribes to elaborate financial intermediation, consultancy fees, revolving-door employment, and campaign contributions in response to increasingly strict antibribery laws illustrates the complexity escalation predicted by part (i) of the theorem (Rose-Ackerman, 1999; Olken and Pande, 2012).

The experience of China's anticorruption campaigns provides a particularly clear illustration. Successive rounds of enforcement have been accompanied by documented increases in the sophistication of corrupt schemes—from direct cash payments to cryptocurrency transfers, from personal favors to multi-layered intermediary networks—without clear evidence of aggregate reduction in corrupt transactions (Svensson, 2005).

The so-called "waterbed effect" in financial regulation—where restrictions in one jurisdiction or instrument class displace activity to less-regulated alternatives—is a direct manifestation of adaptive morphology in the specific context of financial corruption (Dimant and Tosato, 2018).

### 5.4 Counterarguments and Responses

#### *5.4.1 Objection: Deterrence Works—Crime Rates Do Respond to Punishment*

The general deterrence literature in criminology finds that punishment severity does affect crime rates for some offense types. Why should corruption be different?

*Response.* The key distinction is between crimes with and without structural drivers. For crimes where the underlying motivation can be addressed by alternatives (e.g., property crime in the presence of welfare systems), deterrence can reduce incidence because the structural driver can be offset. For corruption, the structural driver is institutional friction, which punishment does not address. Deterrence theory itself recognizes that the certainty of punishment matters more than its severity (Becker, 1968), and the adaptive morphology of corruption acts precisely by reducing the certainty of detection (decreasing $p_d$ through increased $\kappa$), thereby neutralizing the deterrent effect of severity increases. The distinction is not that corruption is somehow immune to incentives; it is that corrupt agents respond to incentives by adapting their methods, a response that is available precisely because corruption operates within institutional structures that provide ample concealment opportunities.

#### *5.4.2 Objection: Singapore and Hong Kong Reduced Corruption Through Strong Enforcement*

These cases are frequently cited as evidence that enforcement can dramatically reduce corruption.

*Response.* Both cases involved simultaneous institutional reform: radical simplification of bureaucratic procedures, creation of transparent digital systems, competitive civil service compensation, and elimination of discretionary bottlenecks. The enforcement component was accompanied by—and arguably secondary to—large reductions in $\Phi$. Attributing the success to enforcement alone confounds the treatment: what was tested was not punishment in isolation but punishment combined with friction reduction. The principle does not deny that enforcement has any effect; it asserts that enforcement without friction reduction induces adaptation rather than reduction, and that the cases typically cited as enforcement successes are in fact cases of combined structural reform.

A note on falsifiability is warranted here. The framework risks unfalsifiable reasoning if successful enforcement is always attributed to simultaneous friction reduction and unsuccessful enforcement always confirms Principle III. To be falsifiable, the framework must specify an ex ante test: identify cases where enforcement severity increased substantially while institutional friction remained constant (no simplification, no automation, no transparency reforms), and observe whether corruption volume decreased. If it did, the principle would be disconfirmed. The available evidence—that enforcement campaigns unaccompanied by structural reform produce temporary reductions followed by adaptation (Persson et al., 2013; Dimant and Tosato, 2018)—is consistent with the principle, but systematic empirical testing with appropriate controls remains a priority for future work.

#### 5.4.3 Objection: The Model Assumes Perfectly Rational Agents

Real agents are boundedly rational and may not optimize concealment complexity perfectly.

*Response.* Bounded rationality weakens the precision of the complexity response but does not eliminate it. Agents need not solve the optimization problem (16) exactly; they need only respond directionally to increased risk by investing more in concealment. This directional response—the qualitative prediction of part (i)—is robust to bounded rationality, satisficing, and trial-and-error learning. Indeed, the empirical evidence for complexity escalation in response to enforcement is overwhelming, suggesting that the directional prediction holds even under realistic cognitive constraints.

## 6 Principle IV: Elimination Through Automation

**Principle 4** (Elimination Through Automation)**.** A necessary condition for any corrupt transaction is the presence of a human agent with personal interests at the point where discretionary authority is exercised. Replacing the human agent with an algorithmic decision-making system at that node removes this necessary condition, thereby eliminating the structural possibility of corruption at that node. Corruption cannot occur where there is no one to corrupt.

The first three principles establish that corruption is structurally generated, self-reinforcing, and resistant to punitive intervention. The fourth principle identifies the most direct structural remedy: remove the human decision-maker from the corruption-susceptible node entirely.

### 6.1 Formal Statement and Proof

**Definition 6.1** (Decision Node)**.** A decision node $d \in D$ in an institutional system is a point at which a discretionary determination is made that affects the allocation of resources, permissions, or opportunities. Each node is characterized by:

- An input set $X_d$ (applications, requests, cases);
- A decision function $f_d : X_d \to Y_d$ mapping inputs to outcomes;
- A decision-maker $a_d \in A$ who implements $f_d$.

**Definition 6.2** (Corruption Susceptibility)**.** A decision node $d$ is corruption-susceptible if and only if:

(C1) **Discretion:** The decision-maker $a_d$ has the authority to choose among multiple outcomes for a given input: $|f_d^{-1}(x)| > 1$ for some $x \in X_d$;

(C2) **Personal interest:** The decision-maker possesses a utility function $u_{a_d} : Y_d \times B \to \mathbb{R}$ that depends on personal benefits $B$ (money, favors, career advancement) in addition to the official outcome;

(C3) **Asymmetric information:** The decision-maker's choice among permissible outcomes is not fully observable by the principal: $\exists y, y' \in Y_d$ such that the principal cannot distinguish whether $f_d(x) = y$ or $f_d(x) = y'$.

These three conditions—discretion, personal interest, and information asymmetry—are jointly necessary for corruption. If there is no discretion, the decision-maker has nothing to sell. If there is no personal interest, the decision-maker has no motivation to deviate. If there is no information asymmetry, any deviation is immediately detected and corrected.

**Theorem 6.3** (Automation Eliminates Node-Level Corruption)**.** *Let $d$ be a corruption-susceptible decision node with human decision-maker $a_d$. Define an algorithmic replacement $\alpha_d$ as a deterministic computational system implementing a fixed decision function $f_d^\alpha : X_d \to Y_d$ with the following properties:*

*(A1) $\alpha_d$ has no utility function over personal benefits: there exists no $u_{\alpha_d}$ that depends on $B$;*

*(A2) $\alpha_d$'s decision function $f_d^\alpha$ is deterministic and auditable: for every input $x$, the output $f_d^\alpha(x)$ is uniquely determined and verifiable.*

*Then the replacement of $a_d$ by $\alpha_d$ eliminates conditions (C2) and (C3) of Definition 6.2, rendering the node corruption-immune.*

*Proof. Elimination of condition (C2).* Condition (C2) requires that the decision-maker possesses a utility function dependent on personal benefits $B$. An algorithmic system $\alpha_d$ is a computational process that maps inputs to outputs according to fixed rules. It does not experience utility, does not receive bribes, does not have family members to favor, does not seek career advancement, and does not fear retaliation. Property (A1) states that no such utility function exists for $\alpha_d$. With (C2) eliminated, the payoff to offering a bribe to $\alpha_d$ is zero, because $\alpha_d$ cannot be influenced by the offer—it has no mechanism for converting a bribe into a change in its decision function.

*Elimination of condition (C3).* Property (A2) states that $\alpha_d$'s decisions are deterministic and auditable. For any input $x$, the output $f_d^\alpha(x)$ can be recomputed by any auditor with access to the algorithm and the input. There is no information asymmetry: the principal can verify every decision by re-executing the algorithm. With (C3) eliminated, even if discretion (C1) were somehow present, any deviation from the prescribed decision rule would be immediately detected.

Since (C2) and (C3) are jointly necessary with (C1) for corruption susceptibility, and both are eliminated by the algorithmic replacement, the node becomes corruption-immune. □

The argument is structural, not technological. It does not depend on the sophistication of the algorithm, on the use of artificial intelligence versus simple rule-based systems, or on any particular computational paradigm. It depends solely on the removal of a human agent with personal interests from the decision point. A simple automated system that approves all applications meeting specified criteria—with no human involvement in the approval process—eliminates corruption at that node as completely as the most advanced AI system would.

### 6.2 Extension: System-Level Corruption Reduction

**Corollary 6.4** (System-Level Corruption Reduction Through Automation)**.** *Let $D_S \subseteq D$ denote the subset of decision nodes amenable to algorithmic replacement. The maximum achievable corruption reduction through automation is:*

$$\Delta\gamma_{\max} = \sum_{d \in D_S} w_d \cdot \gamma_d \tag{27}$$

*where $\gamma_d$ is the corruption rate at node $d$ and $w_d$ is the weight of node $d$ in aggregate corruption (measured by transaction volume, resource allocation, or social impact).*

Not all decision nodes are amenable to automation. Judicial sentencing, diplomatic negotiation, and emergency response involve judgment that current algorithmic systems cannot replicate. The corollary identifies automation as a partial but structurally guaranteed solution: every node that can be automated is a node at which corruption is permanently eliminated.

### 6.3 The Design-Level Corruption Problem

The most important counterargument to the automation principle concerns corruption at the design stage: if humans design the algorithms, the corruption may be embedded in the code rather than occurring at the point of execution.

**Proposition 6.5** (Design-Level Corruption Is Qualitatively Different)**.** *Corruption in the design of an algorithmic system differs from corruption at the point of execution in three structural respects:*

(i) ***Temporality:*** *Design-level corruption occurs once, at the time of system specification. Execution-level corruption recurs with every transaction.*

(ii) ***Detectability:*** *Algorithmic decision rules are inspectable, testable, and auditable. A biased algorithm can be detected through statistical analysis of its outputs—differential approval rates, anomalous patterns, systematic deviations from stated criteria—in a way that a corrupt human decision-maker's biases cannot.*

(iii) ***Scalability of correction:*** *Correcting a biased algorithm requires modifying code and redeploying the system—a single action that corrects all future decisions simultaneously. Correcting a corrupt human requires identifying the individual, proving the corruption, navigating legal proceedings, and finding a replacement, all of which must be repeated for each corrupt actor.*

*Proof. Part (i).* By definition, the algorithm $f_d^\alpha$ is specified at design time and then applied uniformly to all subsequent inputs. Any bias introduced at design time is fixed—it does not require ongoing corrupt transactions to maintain. In contrast, human corruption at the execution level requires a separate corrupt act for each transaction. The frequency of corrupt events under automation is therefore bounded by the number of design iterations (finite and small), while the frequency under human decision-making is proportional to the number of transactions (potentially unbounded).

*Part (ii).* The determinism of $f_d^\alpha$ (property A2) means that the algorithm's behavior can be tested against a reference dataset of inputs with known correct outputs. Statistical tests—disparate impact analysis, bias audits, fairness metrics—can detect systematic deviations with high power. For a human decision-maker, equivalent detection requires monitoring individual decisions over time, a process that is slower, noisier, and complicated by the agent's ability to vary their behavior to avoid detection (the adaptive morphology of Principle III). The detection probability for design-level corruption satisfies:

$$p_d^{\text{design}} > p_d^{\text{execution}}$$

because algorithmic outputs are deterministic, recordable, and testable at arbitrary scale.

*Part (iii).* Correcting a biased algorithm is a software deployment: modify the code, test, deploy. The cost is $O(1)$ regardless of the number of future transactions. Correcting human corruption requires per-instance intervention: each corrupt actor must be individually identified, investigated, prosecuted, and replaced, at cost $O(n)$ where $n$ is the number of corrupt actors. □

Design-level corruption is therefore a real but qualitatively more tractable problem than execution-level corruption. It is temporally bounded, more detectable, and correctable at system scale. The appropriate response is not to reject automation but to implement robust algorithmic auditing, open-source decision rules where possible, and adversarial testing of deployed systems.

### 6.4 Empirical Support

The automation of tax collection in many countries provides direct evidence. Where tax assessments are computed algorithmically from declared income and assets, with human involvement limited to auditing outliers, corruption in routine tax administration has been dramatically reduced (Olken, 2007). Similarly, automated procurement systems that match bids to specifications without human intermediation have reduced procurement corruption in multiple jurisdictions (Rothstein, 2011).

Estonia's e-governance system, which automates the vast majority of citizen–government interactions, has been associated with consistently low corruption levels. The system eliminates human decision-making at thousands of routine administrative nodes, directly implementing the mechanism described in Theorem 6.3.

Conversely, domains where automation has not been implemented—zoning decisions, judicial proceedings, diplomatic appointments—remain corruption-susceptible, consistent with the prediction that corruption persists wherever a human agent with personal interests exercises discretionary authority.

### 6.5 Counterarguments and Responses

#### *6.5.1 Objection: Algorithms Can Be Biased, Opaque, and Harmful*

The algorithmic fairness literature has extensively documented cases where automated systems perpetuate or amplify biases, raising the concern that automation merely replaces one form of injustice with another.

*Response.* The relationship between algorithmic bias and corruption is more nuanced than a clean separation suggests. In the paradigmatic case, they are distinct: bias arises from training data or design choices that systematically disadvantage certain groups, while corruption requires a human agent who personally benefits from deviating from prescribed criteria. An algorithm that unfairly rejects loan applications from a demographic group due to biased training data is biased but not corrupt—no one receives bribes or favors in exchange for the rejections.

However, the boundary becomes blurred in cases where design choices are themselves influenced by corrupt interests. When a procurement algorithm is designed with weighting criteria that systematically favor politically connected firms, the result is simultaneously bias (systematic disadvantage to non-connected firms) and corruption (designers benefited from the design choice). In such cases, the corruption has migrated from the execution level to the design level, which is addressed by Proposition 6.5.

The critical distinction is practical: execution-level corruption (a human official taking bribes on each transaction) recurs with every decision and is difficult to detect because each act is individual and variable. Design-level corruption (embedding favoritism in code) occurs once and then produces detectable statistical patterns in outputs. The remedy for design-level corruption—algorithmic auditing, output monitoring, open-source code review—is structurally different from and more tractable than the remedy for execution-level corruption. Automation does not eliminate all forms of institutional injustice; it transforms corruption from a recurring, distributed, hard-to-detect problem into a one-time, concentrated, auditable one.

### 6.5.2 Objection: The Political Economy of Automation Is Itself Corruptible

The choice of which nodes to automate, which criteria to encode, and who audits the automated systems are themselves discretionary decisions subject to corruption. Automation may simply shift the corruption to a higher level of institutional architecture.

*Response.* This objection is substantially correct, and the framework must acknowledge it explicitly. The decisions about what to automate, how to specify decision criteria, and who controls the auditing process are indeed discretionary and corruption-susceptible. Principle IV does not eliminate corruption from the institutional system as a whole; it eliminates corruption at the automated nodes and pushes residual corruption upward to the design and governance layers.

The argument for automation rests on two structural observations. First, the number of design-level decisions is orders of magnitude smaller than the number of execution-level transactions. If a procurement system processes 100,000 transactions per year, execution-level corruption requires 100,000 corrupt acts; design-level corruption requires corrupting a single design specification. The attack surface shrinks dramatically even if the corruption itself is not eliminated. Second, design-level decisions are more amenable to transparency, public scrutiny, and formal auditing than execution-level decisions, because they produce documented artifacts (code, specifications, criteria) rather than ephemeral acts (conversations, cash transfers). These structural advantages do not guarantee corruption-free automation, but they guarantee that corruption under automation is a smaller and more tractable problem than corruption without it.

#### 6.5.3 *Objection: Automation Eliminates Human Judgment Where It Is Needed*

Some decisions require contextual understanding, empathy, or moral reasoning that algorithms lack.

*Response.* The principle does not advocate replacing all human decision-making with algorithms. It identifies a specific class of decisions—routine, criteria-based determinations at corruption-susceptible nodes—where algorithmic replacement is both feasible and corruption-eliminating. Where genuine human judgment is required (judicial discretion, complex policy trade-offs, crisis response), humans must remain in the loop, and other anticorruption mechanisms (transparency, accountability, structural reform per Principles I–III) must be applied. The principle applies to the substantial share of institutional decisions that are routine, rule-governed, and susceptible to corruption precisely because they involve human discretion where none is structurally necessary.

# 7 The Totalizing Ideal Constraint

The four principles developed above might suggest an unqualified prescription: reduce institutional friction to zero, automate all decision nodes, and eliminate corruption entirely. This section demonstrates that such a prescription is not merely impractical but structurally self-defeating. The complete eradication of all informal adaptive behavior—including the residual, low-level practices that function as institutional flexibility—constitutes a totalizing optimization that destroys the adaptive variance required for institutional viability under irreducible uncertainty.

This result draws on the formally proved Asymmetry of Totalizing Ideals (Kriger, 2025), which establishes a general structural law for complex adaptive systems. We apply that result to the specific context of institutional corruption, demonstrating that anticorruption reform is subject to a fundamental constraint: the pursuit of absolute institutional purity generates greater long-term systemic damage than the maintenance of bounded imperfection.

## 7.1 Institutional Systems as Complex Adaptive Systems

An institutional system satisfies the defining properties of a complex adaptive system (CAS): it comprises multiple interacting agents; it exhibits emergent behavior not predictable from individual components; it operates under irreducible uncertainty about the environment; and its long-term viability depends on its capacity to adapt to changing conditions.

**Assumption 7.1** (Irreducible Institutional Uncertainty)**.** No institutional design can fully anticipate all future contingencies. The true parameter $\theta^*$ governing the optimal institutional configuration is unknown and cannot be determined from any finite sequence of observations:

$$H(\theta^*|o_1, \ldots, o_n) > 0 \quad \forall n < \infty \tag{28}$$

where $H$ denotes Shannon entropy and $\{o_1, \ldots, o_n\}$ is any finite observation sequence.

This assumption is not controversial: it merely formalizes the recognition that institutional designers cannot foresee all future circumstances, that the environment in which institutions operate changes in unpredictable ways, and that the optimal institutional configuration shifts with the environment. Any claim that a particular institutional design is permanently optimal for all future conditions would require infinite knowledge of future contingencies, which is excluded by assumption.

### 7.2 Totalizing Anticorruption as Variance Destruction

A totalizing anticorruption regime is one that seeks to eliminate all deviation from prescribed institutional behavior, including the residual informal practices that provide institutional flexibility.

**Definition 7.2** (Totalizing Anticorruption Regime)**.** An anticorruption regime $R_T$ is totalizing if it satisfies:

(T1) All deviations from prescribed procedures are classified as corruption to be eliminated;

(T2) The regime actively minimizes behavioral variance, driving institutional behavior toward a single prescribed pattern;

(T3) No distinction is made between harmful corruption and adaptive informal behavior;

(T4) Signals indicating that formal rules may be suboptimal are treated as evidence of corruption rather than as information about institutional design.

The parallel with the general definition of totalizing optimization (Kriger, 2025) is precise. Condition (T1) corresponds to error reclassification—treating all variance as defect. Condition (T2) corresponds to variance minimization. Condition (T4) corresponds to signal suppression—discounting corrective feedback.

### 7.3 The Asymmetry Result Applied to Institutional Corruption

**Theorem 7.3** (Anticorruption Totalizing Asymmetry)**.** *Let $\mathcal{I}$ be an institutional system satisfying Assumption 7.1. Let $R_T$ be a totalizing anticorruption regime (Definition 7.2) and let $R_V$ be a variance-preserving regime that maintains institutional flexibility $V_t \geq V_{\min} > 0$. Then for the expected long-term institutional damage:*

$$D(\mathcal{I}, R_T) > D(\mathcal{I}, R_V) \tag{29}$$

*provided the initial gap between institutional design and optimal design exceeds the system's learning capacity before variance collapses.*

*Proof.* The proof follows the structure of the general Asymmetry of Totalizing Ideals (Kriger, 2025), instantiated for the institutional context.

*Step 1: Variance dynamics under $R_T$.* Under a totalizing anticorruption regime, all behavioral variation is treated as corruption and suppressed. The adaptive variance, which measures the information that institutional diversity carries about the unknown optimal configuration, decays exponentially:

$$V_t^T \leq V_0 \cdot e^{-\alpha t} \tag{30}$$

where $\alpha > 0$ is the enforcement intensity of the totalizing regime.

*Step 2: Loss of learning capacity.* By the information-theoretic learning bound (Cover and Thomas, 2006), the rate at which the system can reduce its design error (the gap between current institutional rules and the rules that would be optimal for the actual environment) is bounded by the adaptive variance:

$$-\frac{d}{dt}\mathbb{E}[\|\hat{\theta}_t - \theta^*\|^2] \leq \kappa \cdot V_t \tag{31}$$

where $\kappa > 0$ is a learning efficiency parameter. Under (30), the total learning capacity is:

$$\kappa \int_0^\infty V_0 e^{-\alpha t}\, dt = \frac{\kappa V_0}{\alpha}$$

If the initial design error exceeds this total capacity—the generic case under Assumption 7.1—the system cannot correct its design before variance is exhausted.

*Step 3: Error accumulation.* Once learning capacity is exhausted, the system operates under a fixed (and incorrect) institutional design. Design errors accumulate without correction:

$$E_t^T = \int_0^t \|\hat{\theta}_\tau - \theta^*\|^2 \, d\tau \geq \delta^2 \cdot t \to \infty \tag{32}$$

where $\delta^2 > 0$ is the residual design error.

*Step 4: Damage divergence.* The institutional damage functional under $R_T$ includes both accumulated design error and the cost of lost adaptability:

$$D(\mathcal{I}, R_T) = \int_0^\infty e^{-\rho t} \left[ c_1 E_t^T + \frac{c_2}{V_t^T + \epsilon} \right] dt$$

*Remark* 7.4 (Justification of the Damage Functional Form). The term $c_2/(V_t + \epsilon)$ requires independent justification, since the theorem's conclusion follows largely from this modeling choice. The functional form captures the following substantive claim: the cost of an institutional system's inability to adapt to environmental change increases without bound as the system's adaptive capacity approaches zero.

Three independent lines of reasoning support this claim.

*Ashby's Law of Requisite Variety* (Ashby, 1956): a regulatory system must possess variety at least matching that of the disturbances it faces. When institutional behavioral variety ($V_t$) falls below environmental variety, the system cannot regulate effectively. The gap between required and available variety creates unaddressed disturbances whose costs accumulate. As $V_t \to 0$, the system becomes incapable of responding to *any* environmental perturbation, and unaddressed costs grow without bound.

*Empirical institutional rigidity costs*: historically, institutional systems that achieved near-total behavioral uniformity—command economies, totalitarian bureaucracies—exhibited catastrophic failure when environmental conditions changed, precisely because they had no internal mechanism for detecting or responding to the mismatch. The costs of such failures (economic collapse, humanitarian crises) were orders of magnitude larger than the costs of the prior institutional imperfections.

*Alternative functional forms*: the qualitative result (divergent damage under totalizing regimes) is robust to the specific functional form. Replacing $c_2/(V_t + \epsilon)$ with $c_2 \cdot V_t^{-\beta}$ for any $\beta > 0$, or with $c_2 \cdot \log(1/V_t)$, produces the same qualitative conclusion: damage diverges as $V_t \to 0$. The result fails only if the cost of lost adaptability is *bounded* as $V_t \to 0$—that is, if there exists some finite cost ceiling beyond which losing additional adaptive capacity causes no further damage. We are not aware of any theoretical or empirical basis for such a ceiling.

We acknowledge that the theorem's conclusion is conditional on this modeling choice: it establishes that *if* the cost of lost adaptability is unbounded as adaptive capacity vanishes (a claim we have argued for but not proven), *then* totalizing regimes produce greater damage than variance-preserving ones.

The second term diverges as $V_t^T \to 0$ (provided the discount rate $\rho < \alpha$), yielding $D(\mathcal{I}, R_T) = \infty$.

*Step 5: Bounded damage under $R_V$.* Under the variance-preserving regime, $V_t^V \geq V_{\min} > 0$, ensuring continuous learning at rate $\kappa V_{\min}$. Design errors are corrected as they arise (within the learning rate bound), and the damage functional converges:

$$D(\mathcal{I}, R_V) = \frac{c_1 C_E}{\rho^2} + \frac{c_2}{V_{\min} \cdot \rho} < \infty$$

The comparison $D(\mathcal{I}, R_T) = \infty > D(\mathcal{I}, R_V) < \infty$ completes the proof. □

### 7.4 What Adaptive Variance Looks Like in Institutional Practice

The mathematical concept of adaptive variance has concrete institutional manifestations. Adaptive variance is preserved when:

- Different jurisdictions experiment with different regulatory approaches, generating information about what works;
- Local officials exercise limited discretion to adapt general rules to local circumstances;
- Informal practices—workarounds, unofficial accommodations, minor procedural shortcuts—provide feedback about where formal rules are misaligned with reality;
- Citizens and firms can choose among multiple channels for interacting with government, creating competitive pressure on institutional performance.

Some of these practices, viewed through a totalizing anticorruption lens, constitute minor corruption or procedural violations. The theorem establishes that eliminating them entirely destroys the institutional system's capacity to detect and correct its own design errors.

### 7.5 The Optimal Level of Anticorruption Enforcement

Theorem 7.3 does not imply that corruption should be tolerated without limit. It implies that the optimal anticorruption regime is one that reduces major, harmful corruption while preserving sufficient institutional flexibility (adaptive variance) for ongoing institutional learning and adaptation.

**Corollary 7.5** (Bounded Optimal Enforcement)**.** *The enforcement intensity* $\alpha^*$ *that minimizes expected long-term institutional damage satisfies:*

$$0 < \alpha^* < \alpha_{total} \tag{33}$$

*where* $\alpha_{total}$ *is the enforcement intensity of a totalizing regime. The optimal enforcement targets high-impact corruption (large allocative distortions, major resource diversions) while tolerating low-impact informal practices that serve as adaptive variance.*

This corollary provides the formal basis for a pragmatic anticorruption strategy: vigorous enforcement against systemic corruption that distorts major institutional functions, combined with tolerance for minor institutional flexibility that serves as the system's mechanism for self-correction and adaptation.

## 7.6 Counterarguments and Responses

### *7.6.1 Objection: Preserved Variance May Be Rent-Seeking, Not Adaptive Signal*

The theorem assumes that the informal behaviors preserved by a variance-preserving regime carry information about institutional design. But the "adaptive variance" preserved might simply be rent-seeking that provides no useful signal. Unlike the immune system, which evolved specifically to learn from pathogen exposure, there is no analogous evolutionary mechanism ensuring that institutional corruption carries useful design information.

*Response.* This is the most serious objection to the totalizing ideal constraint, and it partially succeeds. The theorem does not guarantee that all preserved variance is informationally useful. Some fraction of informal behavior is pure rent-seeking with no diagnostic value. The theorem establishes only that *eliminating all* variance is structurally damaging, not that *all* variance is beneficial.

The resolution lies in recognizing two distinct components of institutional informality: (a) *diagnostic informality*—workarounds that reveal misalignment between formal rules and legitimate needs, providing actionable feedback for institutional redesign—and (b) *parasitic informality*—rent-seeking that exploits institutional weakness without generating useful information. The optimal variance-preserving regime would selectively maintain (a) while suppressing (b). In practice, this distinction is imperfect, but it can be approximated through institutional feedback mechanisms: when informal behavior is accompanied by complaints about specific rules, requests for procedure changes, or patterns of systematic circumvention of particular regulations, it carries diagnostic value. When it consists solely of resource extraction with no identifiable systemic signal, it is parasitic.

The immune system analogy is therefore more apt than it initially appears, but for a different reason. The immune system does not benefit from all pathogens; it benefits from exposure to a controlled range that builds adaptive capacity. Similarly, institutions do not benefit from all corruption; they benefit from maintaining the flexibility to detect and respond to misalignment between rules and reality. The theorem's prescription is not "tolerate all corruption" but "do not pursue institutional uniformity so aggressively that the system loses its capacity to detect its own design errors."

### *7.6.2 Objection: This Is a License for Corruption*

Arguing that some corruption should be tolerated is morally and politically untenable.

*Response.* The result does not license corruption. It constrains the form that effective anticorruption policy can take. The theorem demonstrates that the specific goal of eliminating *all* informal behavior produces worse outcomes than targeting *major* corruption while maintaining institutional flexibility. This is analogous to the recognition in medicine that a zero-pathogen environment (a perfectly sterile body) is lethal because the immune system requires exposure to function. The prescription is not "accept infection" but "maintain the conditions for immune function while targeting dangerous pathogens." Translated to institutional terms: target structural corruption aggressively while preserving the adaptive variance that allows institutions to learn and self-correct.

### *7.6.3 Objection: The Distinction Between Harmful Corruption and Adaptive Behavior Is Impossible to Draw*

In practice, no clear line separates "minor institutional flexibility" from "low-level corruption," making the theorem's prescription unimplementable.

*Response.* The difficulty is real but does not invalidate the principle. Several operational criteria can guide the distinction: (a) magnitude of resource diversion (large diversions are corruption; minor procedural shortcuts are flexibility); (b) directionality of benefit (practices that benefit only the official are corruption; practices that benefit the citizen or improve service delivery are flexibility); (c) systemic effect (practices that erode institutional trust are corruption; practices that maintain institutional function are flexibility). These criteria are imperfect, but they provide a better basis for policy than the totalizing alternative, which treats all deviation identically and thereby destroys institutional adaptive capacity.

#### *7.6.4 Objection: The Mathematical Model Is Too Abstract to Guide Policy*

The theorem uses information-theoretic concepts (adaptive variance, mutual information, Shannon entropy) that are far removed from the practical realities of governance.

*Response.* The mathematical abstraction serves to establish the structural impossibility result with rigor. The practical implication is concrete: anticorruption policy should focus on reducing institutional friction (Principle I), automating corruption-susceptible decision nodes (Principle IV), and targeting major corruption, rather than pursuing the elimination of all informal behavior. These are actionable prescriptions that follow from the theorem without requiring policymakers to compute Shannon entropies.

## 8 Toy Models and Simulations

This section presents computational toy models illustrating the dynamics described by each principle. These simulations are intended as qualitative demonstrations, not as empirical calibrations. The code is provided in Appendix A.

### 8.1 Model 1: Friction–Corruption Monotonicity (Principle I)

The first simulation implements replicator dynamics on a population of $N = 1000$ agents choosing between compliant and corrupt strategies. The payoff structure incorporates a benefit $b = 3.0$ for institutional cooperation, an essential compliance cost $c_E = 0.5$, a detection probability $p_d = 0.3$, and a penalty $F = 2.0$. The gain from corruption $\delta$ and the discretionary compliance cost $c_D$ both increase linearly with institutional friction $\Phi$.

Figure 1 displays the equilibrium corruption rate $\gamma$ as a function of $\Phi$. The relationship is monotonically increasing, as predicted by Theorem 3.4. At zero friction, corruption is negligible (the cooperative equilibrium dominates). As friction increases, the equilibrium corruption rate rises continuously, passing through a critical threshold $\Phi^*$ at which the majority of agents have adopted corrupt strategies. The shape of the curve—concave at low friction, steepening at intermediate levels, and saturating near full corruption at high friction—reflects the interaction between the structural incentive $\Delta(\Phi)$ and the population distribution of moral costs $\mu_i$.

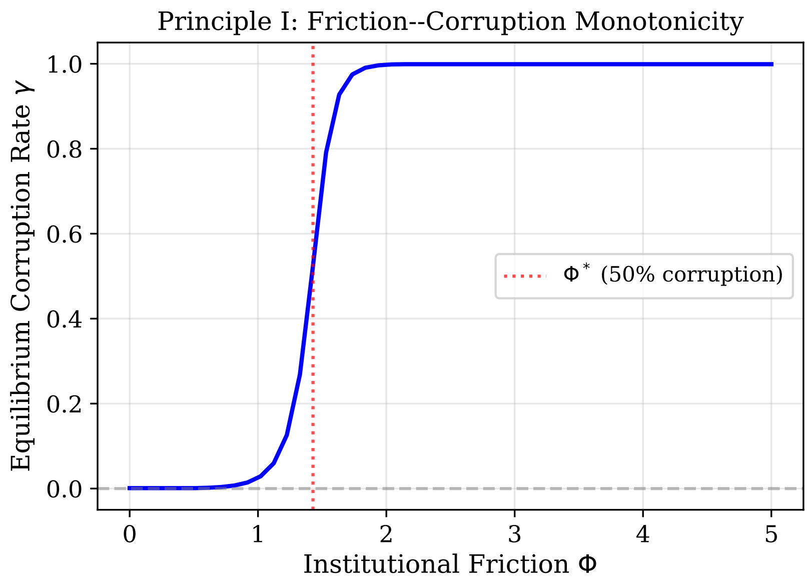


Figure 1: Equilibrium corruption rate as a function of institutional friction. The dotted red line marks the threshold at which corruption exceeds 50%. Below this threshold, compliance is the majority strategy; above it, corruption dominates. The monotonic relationship confirms Theorem 3.4.

### 8.2 Model 2: Selection Dynamics (Principle II)

The second simulation illustrates the displacement of compliant agents under competitive selection. Two panels compare the dynamics below and above the critical friction threshold.

In panel (a) of Figure 2, with $\Phi < \Phi^*$, the fraction of corrupt agents fluctuates around a moderate level. The advantage of corruption is small, and stochastic variation prevents systematic displacement of compliant agents. Multiple simulation runs (shown as transparent traces) exhibit diverse trajectories, indicating that the system does not strongly select for or against corruption.

In panel (b), with $\Phi > \Phi^*$, the dynamics are qualitatively different. All simulation runs converge toward full corruption, regardless of the initial fraction of corrupt agents. The competitive advantage of corruption is large enough to overwhelm stochastic variation, producing systematic displacement of compliant agents. This is the selection dynamics predicted by Theorem 4.4: compliant agents are outcompeted and removed from the population of institutional actors.

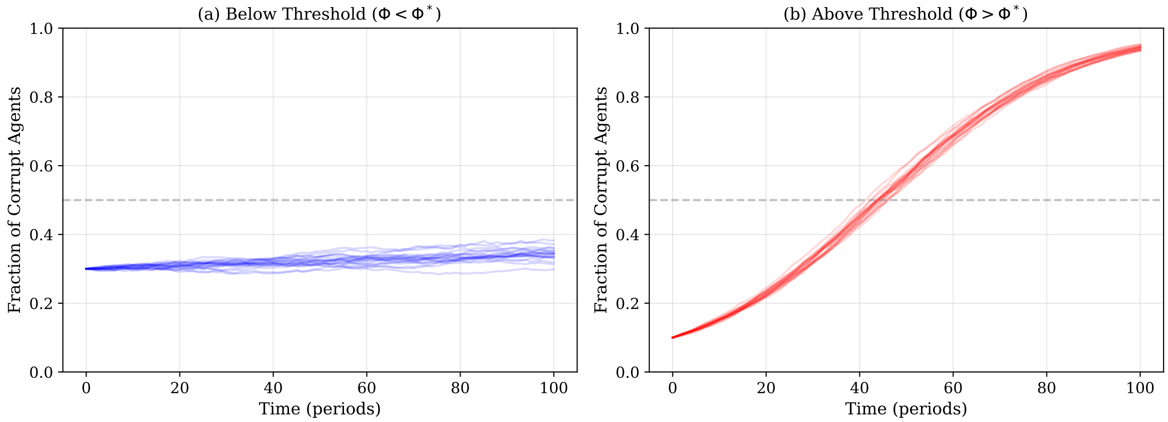


Figure 2: Selection dynamics below and above the critical friction threshold. Each trace represents one simulation run ($n = 20$ runs per panel). (a) Below threshold: diverse outcomes, no systematic displacement. (b) Above threshold: all runs converge to full corruption, illustrating the displacement mechanism of Theorem 4.4.

### 8.3 Model 3: Enforcement–Concealment Arms Race (Principle III)

The third simulation models the response of rational corrupt agents to increasing penalty severity $F$. The detection probability follows $p_d(\kappa) = p_{\max}e^{-\beta\kappa}$ with $p_{\max} = 0.8$ and $\beta = 0.5$. The concealment cost is quadratic: $c_\kappa(\kappa) = \eta\kappa^2$ with $\eta = 0.1$. The structural advantage of corruption is $\Delta = 5.0$.

Figure 3 displays four aspects of the system's response to increasing penalties:

Panel (a) shows that optimal concealment complexity $\kappa^*$ increases monotonically with penalty severity, as proved in Theorem 5.3(i). Harsher penalties induce more sophisticated corruption, not less corruption.

Panel (b) shows the corresponding decline in detection probability. As agents invest in more complex concealment, the probability of catching them falls toward zero, neutralizing the deterrent effect of the increased penalties.

Panel (c) decomposes the total social cost into concealment waste and enforcement expenditure. Both components increase with penalty severity, and their sum—the total social cost—rises monotonically, confirming Theorem 5.3(iii).

Panel (d) shows that the volume of corruption (the fraction of agents choosing corrupt strategies) remains approximately constant across the range of penalty severities, confirming Theorem 5.3(ii). The system's response to enforcement is adaptation, not deterrence.

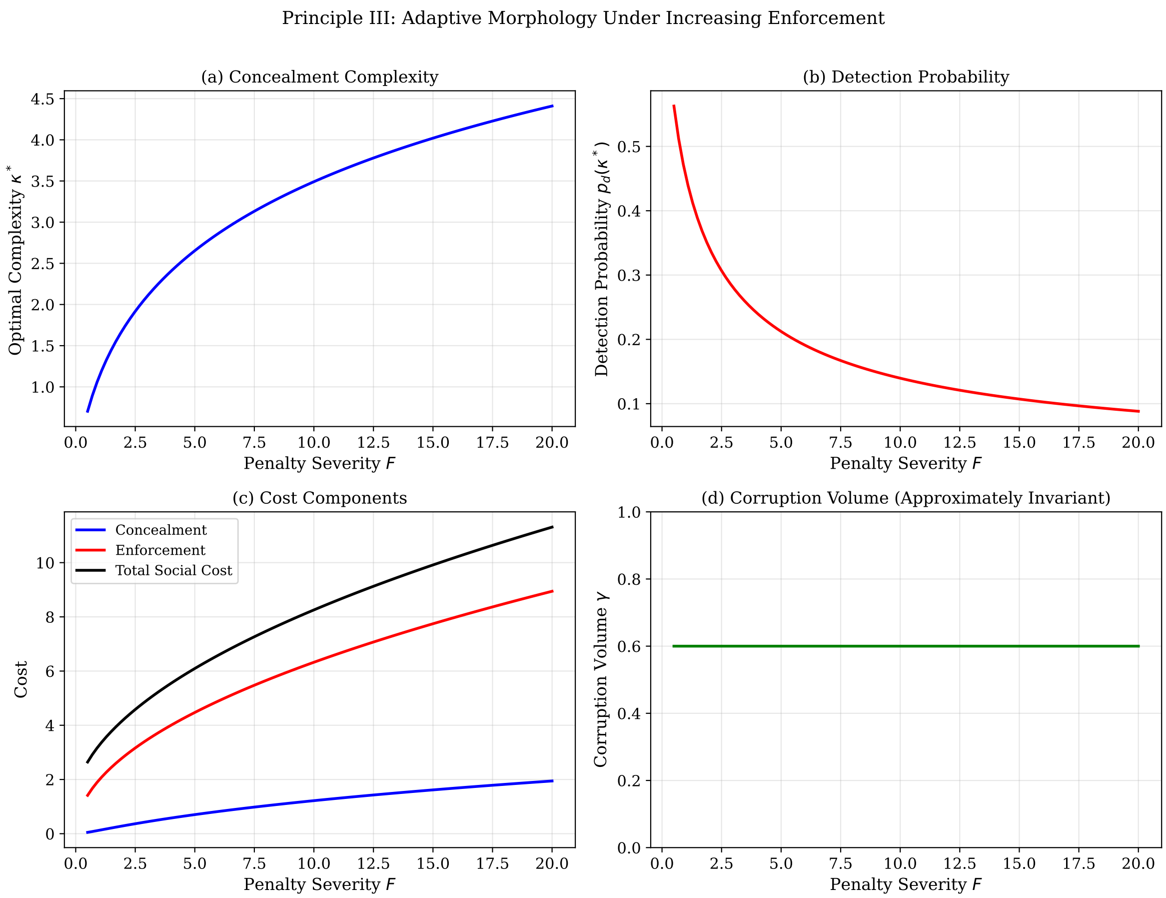


Figure 3: Adaptive morphology under increasing enforcement. (a) Concealment complexity rises with penalty severity. (b) Detection probability falls as concealment improves. (c) Total social cost (concealment + enforcement) increases monotonically. (d) Corruption volume remains approximately invariant. These patterns confirm the three parts of Theorem 5.3.

### 8.4 Model 4: Corruption Elimination Through Automation (Principle IV)

The fourth simulation models a system with $N = 50$ decision nodes, each with a heterogeneous corruption rate drawn from a Beta distribution. Nodes are progressively automated, starting with the most corruption-prone.

Panel (a) of Figure 4 shows the aggregate weighted corruption rate as a function of the percentage of nodes automated. The decline is steepest at low automation levels (automating the worst nodes first yields the largest gains) and continues monotonically toward zero as full automation is achieved.

Panel (b) provides a node-level comparison for the 15 most corrupt nodes, showing their corruption rates before and after automation. The effect is categorical: automation reduces each node's corruption rate to zero, consistent with Theorem 6.3. There is no partial effect, no gradual reduction, no adaptation—once the human decision-maker is removed, the structural possibility of corruption at that node is eliminated.

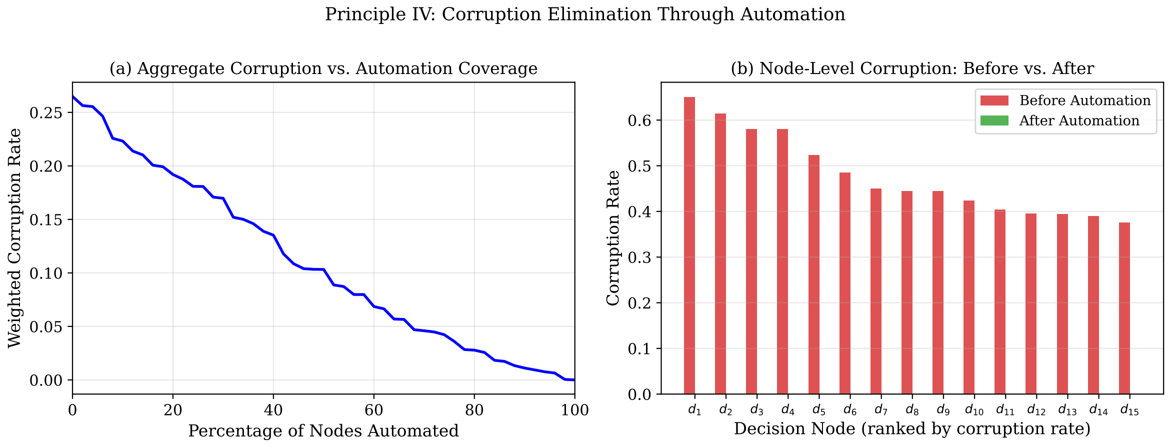


Figure 4: Corruption elimination through node automation. (a) Aggregate corruption declines monotonically as nodes are automated. (b) Node-level comparison showing complete corruption elimination at automated nodes.

### 8.5 Model 5: Totalizing Ideal Constraint

The fifth simulation compares two anticorruption regimes: a totalizing regime that drives adaptive variance to zero, and a variance-preserving regime that maintains a minimum level of institutional flexibility.

Figure 5 displays four aspects of the comparison:

Panel (a) shows adaptive variance over time. Under the totalizing regime (red), variance decays exponentially toward zero. Under the variance-preserving regime (blue), variance stabilizes at $V_{\min} = 0.3$.

Panel (b) tracks institutional design error. Under the totalizing regime, error initially decreases (while variance remains positive and learning is possible) but then stabilizes at a persistent residual level as learning capacity is exhausted. Under the variance-preserving regime, error decreases continuously, reaching a lower long-term level because learning capacity is maintained.

Panel (c) shows instantaneous loss. The totalizing regime's loss diverges as variance approaches zero (the cost of lost adaptability becomes unbounded), while the variance-preserving regime's loss remains bounded.

Panel (d) shows cumulative damage. The totalizing regime's damage grows without bound, while the variance-preserving regime's damage converges to a finite value. This confirms the central result of Theorem 7.3: totalizing anticorruption produces greater long-term institutional damage than bounded, variance-preserving reform.

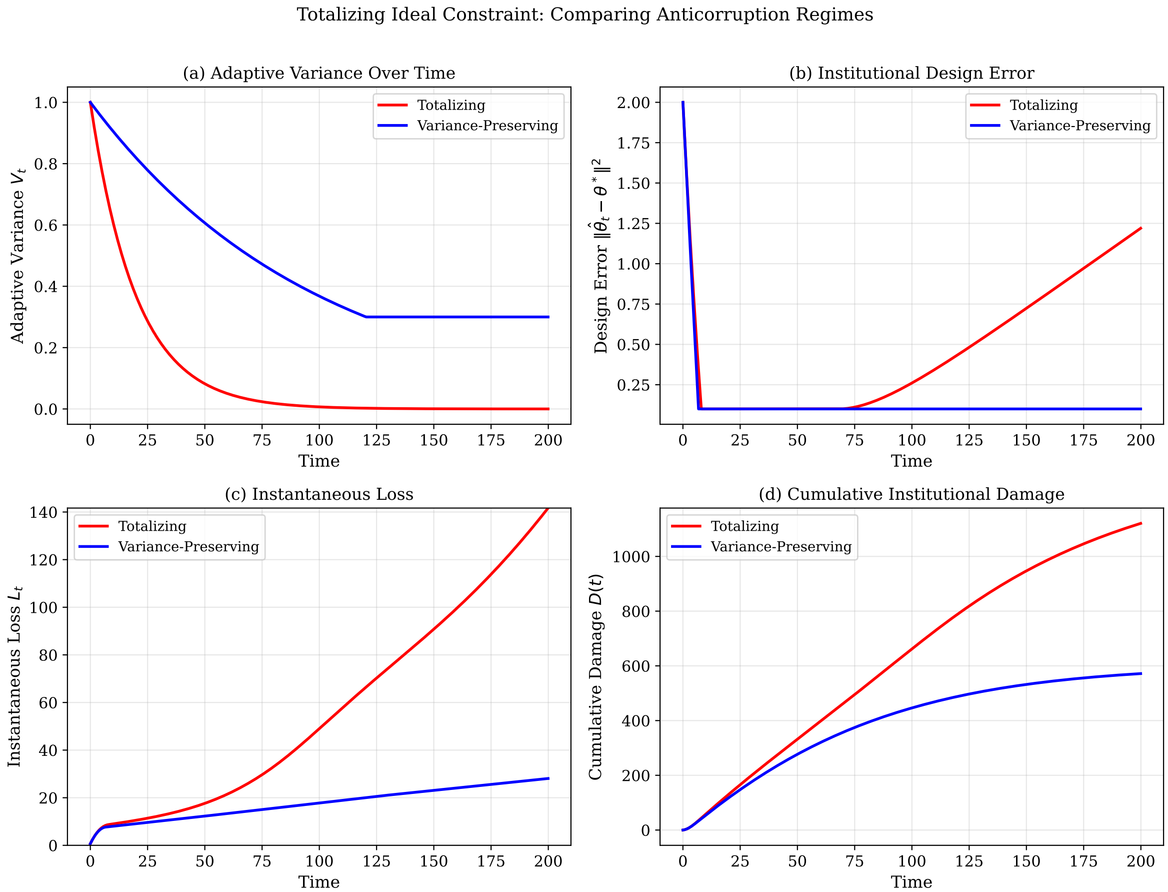


Figure 5: Comparison of totalizing (red) and variance-preserving (blue) anticorruption regimes. (a) Adaptive variance. (b) Institutional design error. (c) Instantaneous loss. (d) Cumulative damage. The totalizing regime produces divergent damage, confirming Theorem 7.3.

# 9 Discussion

## 9.1 Relation to Existing Literature

The framework developed in this paper intersects with several research traditions and clarifies their interrelationships.

The "grease the wheels" hypothesis (Leff, 1964; Huntington, 1968)—that corruption can improve efficiency in dysfunctional institutions—finds partial support in Principle I but is fundamentally reinterpreted. The efficiency gain attributed to corruption is more precisely attributed to the circumvention of unnecessary institutional friction. The policy implication is not that corruption should be tolerated as beneficial but that the friction generating it should be removed. When friction is eliminated, the efficiency gain is achieved without the corrosive side effects of corruption. Empirical tests of the grease hypothesis (Méon and Weill, 2010; Aidt, 2009) that find ambiguous results are consistent with this reinterpretation: corruption appears beneficial in high-friction environments (because it circumvents friction) and harmful in low-friction environments (because it introduces distortions without offsetting benefit).

The collective action perspective on corruption (Persson et al., 2013; Mungiu-Pippidi, 2015) argues that corruption persists because no individual agent has an incentive to deviate from a corrupt equilibrium. Principle II formalizes this insight and extends it: the collective action failure is not a spontaneous coordination problem but a structurally imposed condition. The critical friction threshold $\Phi^*$ identifies the point at which the system transitions from a regime where individual moral agency can influence outcomes to one where structural forces dominate. This provides a formal basis for the claim that anticorruption strategies must be "systemic" rather than "individualistic" (Rothstein, 2011).

The transaction-cost approach to institutions (North, 1990; Williamson, 1985) provides the economic foundation for the concept of institutional friction. The present framework contributes a specific mechanism by which transaction costs generate corruption: when the costs of compliance with formal rules exceed the costs of circumvention (adjusted for detection risk and moral cost), corruption becomes the economically rational strategy. This connects the abstract concept of transaction costs to the specific phenomenon of corruption through a formally specified payoff structure.

The deterrence literature (Becker, 1968) models crime as a rational choice balancing expected gains against expected penalties. Principle III extends this model by incorporating the agent's ability to invest in concealment. Standard deterrence theory treats detection probability as exogenous; our framework endogenizes it, showing that agents respond to increased penalties not only by reducing criminal activity but also by increasing concealment investment. This endogenous detection probability is the mechanism through which the adaptive morphology of corruption operates.

The principal-agent approach to corruption and organizational design (Tirole, 1986; Banerjee, 1997) models corruption as collusion between agents in a hierarchy, where supervisors and subordinates can conspire against the principal. This literature addresses mechanisms closely related to our framework—particularly the role of institutional design in enabling or constraining collusion—through a different formalism. Our contribution is complementary: where the principal-agent models derive optimal contracts and monitoring structures for a given organizational hierarchy, our framework asks the prior question of why the hierarchy generates corruption at all, locating the answer in institutional friction rather than in the information structure of the hierarchy alone. A synthesis of the two approaches—using the principal-agent framework's precision about monitoring and contracts within our framework's structural analysis of friction-driven perturbation—is a promising direction for future work.

#### *9.1.1 A Note on Self-Citation and Foundational Dependencies*

The totalizing ideal constraint (Section 7) relies on the Asymmetry of Totalizing Ideals (Kriger, 2025), and the formal apparatus of perturbation analysis draws on the unified theory of self-organizing systems (Kriger, 2026). Both are working papers by the present author that have not yet been independently reviewed and published in peer-reviewed venues. The dependent results in this paper therefore inherit the uncertainty of those foundations. We have presented the relevant arguments self-containedly where possible—in particular, the proof of Theorem 7.3 is reproduced in full rather than cited by reference—so that readers can evaluate the reasoning on its own merits. Both working papers are available from the author upon request and will be deposited in an open-access repository concurrent with this paper's publication. If the foundational results are found to contain errors, the claims that depend on them (particularly Theorem 7.3) would require re-examination; however, the first three principles are independent of these foundations and stand on their own.

### 9.2 Falsifiability Criteria

Each principle generates specific empirical predictions that can, in principle, be tested. We specify time horizons and measurement strategies to make these predictions operationally falsifiable.

*Principle I* predicts that reducing institutional friction (holding enforcement constant) will reduce corruption. A controlled policy experiment—simplifying business registration procedures in a randomly selected subset of jurisdictions while maintaining the same enforcement regime—would test this. The predicted effect should be observable within 2–5 years (the timescale for behavioral equilibration in institutional settings). Corruption can be measured through multiple independent proxies: bribery prevalence surveys, discrepancies between official and actual processing times, and audit-detected irregularities. If corruption rates do not decrease in the simplified jurisdictions over this period, the principle is disconfirmed.

*Principle II* predicts that replacing personnel without structural reform will not reduce corruption. The model predicts that corruption will return to pre-replacement levels within one competitive cycle—the time required for selection dynamics to filter the replacement cohort. In bureaucratic contexts with annual performance reviews and competitive promotions, this corresponds to approximately 3–7 years. If a jurisdiction replaces all officials at corruption-prone agencies with new personnel selected for high integrity, and corruption measured by the same proxies remains permanently reduced beyond this period without accompanying structural reform, the principle is disconfirmed.

*Principle III* predicts that increasing enforcement severity (holding friction constant) will increase concealment sophistication without reducing corruption volume. This presents a fundamental measurement challenge: "corruption volume" includes undetected corruption, which is by definition unobservable. Indirect measurement strategies include: (a) average complexity of detected schemes (predicted to increase); (b) time lag between corrupt act and detection (predicted to increase); (c) ratio of whistleblower-detected to audit-detected corruption (predicted to increase, as sophisticated schemes evade audits but remain visible to insiders); (d) economic proxies such as unexplained wealth accumulation or discrepancies between official income and lifestyle among officials. If enforcement increases coincide with simpler detected schemes, shorter detection lags, and lower proxy-based corruption estimates, the principle is disconfirmed.

*Principle IV* predicts that automating a decision node will reduce corruption at that node to zero. The prediction is categorical and therefore strongly falsifiable: if corruption persists at a fully automated node (through manipulation of inputs, exploitation of edge cases, or other mechanisms not involving human discretion at the node itself), the principle is disconfirmed for that node type. The measurement is straightforward: compare corruption indicators (bribery reports, processing irregularities, favoritism patterns) at the node before and after automation.

## 9.3 Policy Implications

The four principles, taken together, suggest a coherent anticorruption strategy that differs substantially from the prevailing emphasis on detection, punishment, and moral reform.

*Primary intervention: friction reduction.* The most effective anticorruption measure is the systematic identification and elimination of unnecessary institutional friction. Every discretionary approval, redundant procedure, and opaque criterion that does not serve a genuine policy objective is a potential corruption node. The policy prescription is straightforward: for each institutional process, determine the minimal set of rules required for the system's stated objectives ($\mathcal{R}_E$), and eliminate or simplify the remainder.

*Secondary intervention: strategic automation.* For decision nodes where human discretion is not structurally necessary, replace human decision-makers with algorithmic systems. This should be pursued in order of corruption impact: automate the nodes with the highest corruption rates and the largest resource allocations first, where the structural guarantee of Theorem 6.3 yields the greatest benefit.

*Tertiary intervention: targeted enforcement.* Enforcement resources should be concentrated on high-impact corruption—major resource diversions, systemic distortions of institutional function—rather than distributed across all forms of institutional irregularity. This targeting is justified by two considerations: (a) Principle III shows that broad enforcement induces costly concealment without reducing corruption volume; (b) the totalizing ideal constraint shows that eliminating all informal behavior is structurally harmful.

*Complementary measure: institutional learning mechanisms.* To maintain the adaptive variance required by the totalizing ideal constraint, institutional systems should incorporate explicit mechanisms for learning from deviations. When citizens or officials circumvent formal procedures, the deviation should be investigated not only as a potential corruption case but also as a signal that the formal procedure may be misaligned with legitimate needs. Systems that treat all deviations as moral failures forgo this diagnostic information.

## 9.4 Limitations and Future Directions

Several limitations should be acknowledged.

### *9.4.1 Endogeneity of Friction*

The framework treats institutional friction as exogenous: friction causes corruption, and the causal arrow runs in one direction. In practice, corrupt officials frequently create unnecessary procedures precisely to generate bribe-extraction opportunities. A customs official who introduces an additional "verification step" that can be bypassed for a fee is simultaneously the product and the producer of institutional friction. This feedback loop—where corruption both results from and generates friction—is not captured by the present model.

Incorporating this endogeneity would require a dynamic model in which the rule set

$\mathcal{R}$ is itself a strategic variable: agents with the authority to modify institutional rules can increase $\Phi$ to expand their corruption opportunities. Such a model would predict that systems above $\Phi^*$ tend to drift toward even higher friction over time (as corrupt agents create new friction to expand their rents), while systems below $\Phi^*$ tend toward lower friction (as compliant agents simplify unnecessary rules). This positive feedback would produce bistability—two stable regimes (low-friction/low-corruption and high-friction/high-corruption) separated by a tipping point—a prediction that is consistent with observed cross-national patterns but that we have not formally derived. This extension is a priority for future work.

### *9.4.2 Scope of Automatable Decision Nodes*

Principle IV's practical impact depends on the size of $D_S$—the set of decision nodes amenable to algorithmic replacement—relative to the total set $D$. The paper does not estimate this ratio, and the omission weakens the principle's policy relevance. Crude estimates are possible: in many bureaucratic systems, the majority of transactions by volume are routine, criteria-based determinations (permit approvals, tax assessments, benefit eligibility, license renewals) that are in principle automatable. The high-impact but low-volume decisions (judicial sentencing, regulatory enforcement prioritization, zoning variance adjudication) typically require genuine human judgment. If 80% of transactions by volume but only 20% of transactions by discretionary impact are automatable, then automation eliminates corruption in the routine majority while leaving the consequential minority untouched. Empirical studies of specific institutional domains are needed to refine these estimates.

### *9.4.3 Context-Dependent Moral Costs*

The moral cost parameter $\mu_i$ is modeled as a fixed characteristic of each agent. Moral psychology research, however, suggests that ethical behavior is highly context-dependent: the same individual may resist corruption in one setting and participate in another. If $\mu_i$ varies with institutional context—as the framework's own logic about structural determinism would suggest—then the population distribution $G(\cdot)$ is itself a function of institutional structure, creating a feedback loop between friction and moral resistance that the model does not capture. Specifically, high-friction environments may erode moral resistance over time (agents who observe universal corruption adjust their moral thresholds downward), while low-friction environments may strengthen it. This would amplify the effects predicted by Principles I and II and create path dependence in institutional corruption, but formalizing this mechanism requires a dynamic model of endogenous moral costs.

### *9.4.4 Selection on the Dependent Variable*

The empirical examples cited throughout the paper are cases where the framework's predictions appear to hold. No cases are examined where the predictions might fail—for instance, jurisdictions where enforcement-heavy approaches appeared to succeed without obvious simultaneous friction reduction, or contexts where friction reduction did not produce expected corruption decreases. If such cases exist, they would provide important boundary conditions; if they do not, their absence should be argued rather than assumed. A systematic empirical survey, including cases that challenge the framework, would substantially strengthen the paper's claims.

### *9.4.5 Interaction Structure Limitations*

The replicator dynamics of Principles I and II assume well-mixed populations; in practice, institutional agents are embedded in networks with heterogeneous structure. Many institutional interactions are one-shot or anonymous (a citizen applying for a permit interacts with whichever clerk is available), which is precisely the setting where the cooperation results cited in Theorem 3.2 do not apply. Assumption 3.1 restricts the theorem's scope to repeated-interaction contexts, but much corruption occurs in one-shot encounters. Extending the framework to cover anonymous, one-shot institutional interactions—where cooperation cannot be sustained by reciprocity—would require different mechanisms (reputation systems, institutional design, monitoring) and is an important direction for future work.

### *9.4.6 Concealment Complexity Bounds*

The adaptive morphology model assumes unbounded concealment technology: $\kappa \in [0, \infty)$. In practice, concealment investment may be bounded by the agent's resources, organizational capacity, or the finite complexity of available financial and legal instruments. A bounded strategy space $\kappa \in [0, \bar{\kappa}]$ would create a detection floor $\underline{p}_d = p_d(\bar{\kappa}) > 0$, below which agents cannot reduce detection probability. Under such a bound, sufficiently severe punishment *would* reduce corruption volume for agents near the participation threshold, partially restoring the deterrence mechanism. The qualitative prediction of complexity escalation (Theorem 5.3(i)) is unaffected by this bound, but the volume invariance claim (part (ii)) would hold only up to the concealment ceiling.

#### *9.4.7 Interpretation of $\theta^*$ for Institutional Systems*

The notation $\theta^*$ (the true unknown parameter governing the environment) is borrowed from statistical learning theory and requires concrete interpretation in the institutional context. For an institutional system, $\theta^*$ represents the configuration of environmental conditions—economic structure, technological capabilities, social norms, demographic patterns, external threats—to which institutional rules should ideally be adapted. No institutional designer has complete knowledge of $\theta^*$, and it shifts over time as society evolves. The adaptive variance $V_t$ measures how much the diversity of institutional behaviors reveals about this shifting target. When institutions permit local experimentation, the resulting variation provides information about which rules work under current conditions and which do not—this is the institutional analog of the informational role of genetic diversity in biological populations.

#### *9.4.8 Future Directions*

Future research might productively extend this framework in several directions: empirical estimation of institutional friction $\Phi$ and the critical threshold $\Phi^*$ in specific institutional contexts; formal modeling of the endogeneity of friction; agent-based simulations incorporating network structure, bounded rationality, and context-dependent moral costs; systematic empirical testing (including cases that challenge the framework's predictions); comparative case studies testing the predictions of Principle III regarding concealment escalation in response to enforcement; estimation of $|D_S|/|D|$ in specific bureaucratic domains; and pilot studies of strategic automation in corruption-susceptible institutional settings.

The framework should also be engaged more systematically with the experimental anticorruption literature—field experiments on monitoring, auditing, and transparency interventions (Olken, 2007)—and with the empirical literature on bureaucratic transfers and corruption (Banerjee, 1997).

## 10 Conclusion

This paper has developed a systems-theoretic framework for understanding public corruption, grounded in four formal principles and constrained by a structural impossibility result.

The **Principle of Systemic Origin** (Section 3) establishes that corruption is a monotonically increasing function of institutional friction. The cooperative, non-corrupt equilibrium is the natural attractor of institutional systems operating under low friction; corruption emerges when the gap between formal requirements and agents' legitimate needs creates structural incentives for circumvention. The policy implication is that reducing institutional friction is the most effective anticorruption intervention.

The **Principle of Systemic Coercion** (Section 4) demonstrates that above a critical friction threshold, corruption becomes the unique individually rational strategy, and competitive selection systematically displaces non-corrupt agents. This explains why anticorruption reforms focused on replacing corrupt individuals with honest ones consistently fail: the replacements face the same structural incentives and are subject to the same selection dynamics.

The **Principle of Adaptive Morphology** (Section 5) proves that increasing punitive enforcement does not reduce the volume of corruption but transforms it into more complex, concealed, and socially costly forms. The enforcement–concealment arms race is a pure deadweight loss whose magnitude is determined by unaddressed institutional friction. The policy implication is that enforcement without structural reform is not merely insufficient but counterproductive.

The **Principle of Elimination Through Automation** (Section 6) identifies the most direct structural remedy for corruption at routine decision nodes: replacing the human decision-maker with an algorithmic system eliminates the necessary condition for corrupt transactions. This is not a probabilistic reduction but a categorical elimination, because an algorithmic system lacks the personal interests that are a prerequisite for corruption.

The **Totalizing Ideal Constraint** (Section 7) establishes the structural boundary on anticorruption ambition: the complete elimination of all informal institutional behavior, including the minor adaptive practices that provide institutional flexibility, produces greater long-term systemic damage than bounded imperfection. Effective anticorruption reform must therefore be targeted rather than totalizing.

The central contribution of the framework is a shift in the explanatory and diagnostic burden. Rather than asking why individuals are corrupt—a question that leads to answers in moral psychology and to interventions targeting individual behavior—the framework asks what structural perturbation makes corruption the rational adaptive response. This question leads to answers in institutional design and to interventions targeting the structure within which individuals operate.

The framework is falsifiable. Each principle generates specific predictions: that friction reduction will reduce corruption (Principle I); that personnel changes without structural reform will not reduce corruption (Principle II); that enforcement escalation will increase concealment complexity without reducing corruption volume (Principle III); that automation of decision nodes will categorically eliminate corruption at those nodes (Principle IV). These predictions can be tested against empirical data and policy experiments.

If the framework is correct, the implications for anticorruption policy are substantial. The dominant approach—monitoring individuals, punishing offenders, and exhorting ethical behavior—addresses symptoms rather than causes. The effective approach, suggested by the framework, is structural: simplify institutions, automate routine decisions, target enforcement at high-impact corruption, and maintain the institutional flexibility required for ongoing adaptation. Corruption is not a disease of the soul. It is a disease of institutions. The remedy is institutional reform, not moral reform.

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## A Simulation Code

The following Python code generates all figures presented in Section 8. The code requires `numpy`, `scipy`, and `matplotlib`.

```
"""
Toy Models and Simulations for:
"Four Principles of Corruption: A Systems-Theoretic Analysis
 of Institutional Failure and Reform"
"""
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import brentq

# ================================================================
# MODEL 1: Friction-Corruption Monotonicity (Principle I)
# ================================================================
def model1_friction_corruption():
    N, T = 1000, 200
    friction_levels = np.linspace(0, 5, 50)
    b, c_E, p_d, F = 3.0, 0.5, 0.3, 2.0
    eq_corruption = []
    for Phi in friction_levels:
        delta = 0.3 * Phi
        c_D = c_E + Phi / N * 10
        payoff_comply = b - c_D
        payoff_corrupt = b + delta - p_d * F - c_E
        x = 0.1
        for t in range(T):
            f_comply = x * (payoff_comply - 0.5) + (1-x) * payoff_comply
            f_corrupt = x * payoff_corrupt + (1-x) * (payoff_corrupt + 0.2)
            f_bar = x * f_corrupt + (1-x) * f_comply
            if abs(f_bar) > 1e-10:
                dx = x * (f_corrupt - f_bar) * 0.1
                x = np.clip(x + dx, 0.001, 0.999)
        eq_corruption.append(x)
    plt.figure(figsize=(6, 4))
    plt.plot(friction_levels, eq_corruption, 'b-', linewidth=2)
    plt.xlabel(r'Institutional Friction $\Phi$')
```

```
    plt.ylabel(r'Equilibrium Corruption Rate $\gamma$')
    plt.title('Principle I: Friction-Corruption Monotonicity')
    plt.grid(True, alpha=0.3)
    plt.savefig('fig1_friction_corruption.pdf')
    plt.close()

# ===============================================================
# MODEL 2: Selection Dynamics (Principle II)
# ===============================================================
def model2_selection_dynamics():
    T, n_runs = 100, 20
    fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))
    for run in range(n_runs):
        frac = [0.3]
        for t in range(T):
            adv = 0.002 + np.random.normal(0, 0.01)
            new = frac[-1] + adv * frac[-1] * (1 - frac[-1])
            frac.append(np.clip(new, 0.01, 0.99))
        axes[0].plot(range(T+1), frac, 'b-', alpha=0.15)
    axes[0].set_title(r'(a) Below Threshold ($\Phi < \Phi^*$)')
    for run in range(n_runs):
        frac = [0.1]
        for t in range(T):
            adv = 0.05 + np.random.normal(0, 0.01)
            new = frac[-1] + adv * frac[-1] * (1 - frac[-1])
            frac.append(np.clip(new, 0.01, 0.99))
        axes[1].plot(range(T+1), frac, 'r-', alpha=0.15)
    axes[1].set_title(r'(b) Above Threshold ($\Phi > \Phi^*$)')
    fig.tight_layout()
    fig.savefig('fig2_selection_dynamics.pdf')
    plt.close()

# ===============================================================
# MODEL 3: Adaptive Morphology (Principle III)
# ===============================================================
def model3_adaptive_morphology():
    F_values = np.linspace(0.5, 20, 100)
    p_max, beta, eta, Delta = 0.8, 0.5, 0.1, 5.0
```

```
    kappa_opt, det_prob, conc_cost = [], [], []
    enf_cost, total_cost, corr_vol = [], [], []
    for F in F_values:
        foc = lambda k: 2*eta*k - beta*p_max*np.exp(-beta*k)*F
        try:
            k_star = brentq(foc, 0, 50) if foc(0)<0 and foc(50)>0 else 0
        except: k_star = 0
        pd_k = p_max * np.exp(-beta * k_star)
        cc = eta * k_star**2
        ec = 2.0 * F**0.5
        pi = Delta - cc - pd_k * F
        gamma = 0.6 if pi > 0 else 0.1
        kappa_opt.append(k_star); det_prob.append(pd_k)
        conc_cost.append(cc); enf_cost.append(ec)
        corr_vol.append(gamma)
        total_cost.append(gamma*2.0 + gamma*cc + ec)
    fig, ax = plt.subplots(2, 2, figsize=(12, 9))
    ax[0,0].plot(F_values, kappa_opt, 'b-', lw=2)
    ax[0,1].plot(F_values, det_prob, 'r-', lw=2)
    ax[1,0].plot(F_values, conc_cost, 'b-', lw=2, label='Concealment')
    ax[1,0].plot(F_values, enf_cost, 'r-', lw=2, label='Enforcement')
    ax[1,0].plot(F_values, total_cost, 'k-', lw=2, label='Total')
    ax[1,0].legend()
    ax[1,1].plot(F_values, corr_vol, 'g-', lw=2)
    fig.tight_layout()
    fig.savefig('fig3_adaptive_morphology.pdf')
    plt.close()

# ==============================================================
# MODEL 4: Automation Effect (Principle IV)
# ==============================================================
def model4_automation():
    np.random.seed(42)
    N_nodes = 50
    corruption = np.random.beta(2, 3, N_nodes) * 0.8
    weights = np.random.exponential(1, N_nodes)
    weights /= weights.sum()
    sort_idx = np.argsort(-corruption)
```

```python
    fracs = np.linspace(0, 1, 51)
    total = []
    for f in fracs:
        n_auto = int(f * N_nodes)
        eff = corruption.copy()
        eff[sort_idx[:n_auto]] = 0.0
        total.append(np.sum(eff * weights))
    fig, ax = plt.subplots(1, 2, figsize=(12, 4.5))
    ax[0].plot(fracs*100, total, 'b-', lw=2)
    ax[0].set_xlabel('Percentage of Nodes Automated')
    n_show = 15
    idx = sort_idx[:n_show]
    x_pos = np.arange(n_show)
    ax[1].bar(x_pos-0.175, corruption[idx], 0.35,
              label='Before', color='#d62728', alpha=0.8)
    ax[1].bar(x_pos+0.175, np.zeros(n_show), 0.35,
              label='After', color='#2ca02c', alpha=0.8)
    ax[1].legend()
    fig.tight_layout()
    fig.savefig('fig4_automation.pdf')
    plt.close()

# ==============================================================
# MODEL 5: Totalizing Ideal Constraint
# ==============================================================
def model5_totalizing():
    T, dt = 200, 0.1
    t = np.arange(0, T, dt)
    alpha, V0, kappa = 0.05, 1.0, 0.3
    rho, c1, c2, eps = 0.02, 1.0, 0.5, 0.01
    V_min, err0, sig = 0.3, 2.0, 0.1
    V_T = V0 * np.exp(-alpha * t)
    err_T = np.zeros_like(t); err_T[0] = err0
    for i in range(1, len(t)):
        err_T[i] = max(err_T[i-1] - kappa*V_T[i]*dt + sig*0.01, 0.1)
    cum_T = np.cumsum(err_T)*dt
    loss_T = c1*cum_T + c2/(V_T + eps)
    dmg_T = np.cumsum(np.exp(-rho*t)*loss_T)*dt
```

```
    V_V = np.maximum(V0*np.exp(-0.01*t), V_min)
    err_V = np.zeros_like(t); err_V[0] = err0
    for i in range(1, len(t)):
        err_V[i] = max(err_V[i-1] - kappa*V_V[i]*dt + sig*0.01, 0.1)
    cum_V = np.cumsum(err_V)*dt
    loss_V = c1*cum_V + c2/(V_V + eps)
    dmg_V = np.cumsum(np.exp(-rho*t)*loss_V)*dt
    fig, ax = plt.subplots(2, 2, figsize=(12, 9))
    ax[0,0].plot(t, V_T, 'r-', lw=2, label='Totalizing')
    ax[0,0].plot(t, V_V, 'b-', lw=2, label='Variance-Preserving')
    ax[0,0].legend()
    ax[0,1].plot(t, err_T, 'r-', lw=2, label='Totalizing')
    ax[0,1].plot(t, err_V, 'b-', lw=2, label='Variance-Preserving')
    ax[0,1].legend()
    ax[1,0].plot(t, loss_T, 'r-', lw=2, label='Totalizing')
    ax[1,0].plot(t, loss_V, 'b-', lw=2, label='Variance-Preserving')
    ax[1,0].set_ylim(0, min(np.max(loss_T), 200))
    ax[1,0].legend()
    ax[1,1].plot(t, dmg_T, 'r-', lw=2, label='Totalizing')
    ax[1,1].plot(t, dmg_V, 'b-', lw=2, label='Variance-Preserving')
    ax[1,1].legend()
    fig.tight_layout()
    fig.savefig('fig5_totalizing.pdf')
    plt.close()

if __name__ == '__main__':
    model1_friction_corruption()
    model2_selection_dynamics()
    model3_adaptive_morphology()
    model4_automation()
    model5_totalizing()
```

## Peer Reviews and Revision History

This section documents the peer review process and the revisions made in response to reviewer feedback. It is maintained as a transparent record of the paper's development.

### First Round

*Summary of Reviewer Feedback*

The reviewer assessed the paper as "ambitious and intellectually stimulating" with "valuable core ideas" but identified several significant concerns requiring major revision. The principal issues were:

1. **Proof rigor.** Several proofs were found to rely on assumptions that do most of the analytical work while being presented as conclusions. Specifically: Theorem 3.1's assumption that $\delta \to 0$ under zero friction was inadequately justified (agents can gain from circumventing essential rules); Theorem 4.1's assumption $\bar{\mu} < \infty$ did not account for the exit option (agents can leave rather than participate); Theorem 5.3(ii)'s volume invariance claim required an unstated scope condition on concealment technology.
2. **Empirical claims.** The empirical sections were criticized as illustrative anecdotes rather than systematic evidence, with insufficient attention to confounding variables (e.g., the Estonia/Georgia example conflates many simultaneous changes with friction reduction).
3. **Falsifiability.** The framework risked unfalsifiable reasoning, particularly regarding the Singapore/Hong Kong cases where successful enforcement was attributed to simultaneous friction reduction.
4. **Automation critique.** Principle IV's treatment of design-level corruption was deemed insufficiently critical. The distinction between algorithmic bias and corruption was called "too clean," and the political economy of automation decisions was not addressed.
5. **Totalizing constraint scope.** The reviewer questioned whether preserved "adaptive variance" actually carries useful institutional information or is merely rent-seeking.
6. **Minor issues.** The essential/discretionary rule distinction needed operationalization; the unused notation $\phi$ in Table 1; heavy self-citation of unpublished working papers; and insufficient engagement with principal-agent models of corruption.

*Revisions Made*

1. **Theorem 3.1 (now with Assumption 3.3).** Introduced explicit Assumption 3.3 (Friction-Dependent Circumvention Gain) acknowledging that $\delta_0 \geq 0$ even under zero friction. The proof now requires $\delta_0 < p_d \cdot F$ as a design condition on essential rules, rather than assuming $\delta \to 0$.
2. **Theorem 4.1 (now with Assumption 4.3).** Added Assumption 4.3 (Bounded Exit) acknowledging that agents can exit the system rather than participate in corruption. The theorem now explicitly applies to the population of agents whose exit costs exceed the gap between moral resistance and structural advantage.
3. **Theorem 5.3(ii).** Weakened claim from strict invariance to approximate invariance. Added Remark 5.4 specifying the scope condition: the result holds when concealment technology is sufficiently rich ($\lim_{\kappa \to \infty} p_d(\kappa) = 0$) and may not hold in contexts with hard detection floors.
4. **Empirical sections.** Relabeled as "Illustrative Empirical Evidence" throughout. Added explicit discussion of confounding variables in the Estonia/Georgia example. Stated clearly that observations are consistent with the framework but do not constitute controlled tests.
5. **Falsifiability.** Added Section 9.2 ("Falsifiability Criteria") specifying testable predictions for each principle. Added falsifiability discussion to the Singapore/Hong Kong response in Section 5.4.2, identifying what specific empirical observations would disconfirm each principle.
6. **Automation critique.** Expanded Section 6.5.1 to acknowledge the blurred boundary between algorithmic bias and design-level corruption. Added new Section 6.5.2 ("The Political Economy of Automation Is Itself Corruptible") directly addressing the objection that automation shifts corruption to higher institutional levels, with structural argument for why this represents a more tractable problem.
7. **Totalizing constraint.** Added new counterargument ("Preserved Variance May Be Rent-Seeking, Not Adaptive Signal") distinguishing diagnostic informality from parasitic informality and acknowledging the objection's partial validity.
8. **Notation.** Removed unused $\phi$ from Table 1. Added $\mu_i$ and $\kappa$ to notation table.
9. **Estimation of $\Phi$.** Added Remark 2.3 to Section 2 discussing operational approaches to estimating institutional friction (comparative benchmarking, functional testing, user-reported burden).
10. **Self-citation.** Added explicit self-citation caveat in Discussion acknowledging that results depending on unpublished working papers inherit their uncertainty.
11. **Principal-agent literature.** Added discussion of Tirole (1986) and Banerjee

(1997) in Discussion, positioning the present framework as complementary to the principal-agent approach.

### Second Round

*Summary of Reviewer Feedback*

The reviewer acknowledged the first-round revisions but identified further concerns requiring major revision. The principal issues were:

1. **Theorem 3.1 interaction structure.** The proof invoked network reciprocity without specifying the interaction network; many institutional interactions are one-shot or anonymous.
2. **Separability of $P$.** The essential/discretionary distinction does heavy lifting; classification uncertainty should be formalized and its downstream effects analyzed.
3. **Practical estimation of $\Phi^*$.** The theorem's applicability depends on real systems exceeding $\Phi^*$, but no method for estimating $\Phi^*$ was provided.
4. **Approximate invariance underspecified.** "Approximately invariant" is not a formal statement; a bound in terms of the moral cost distribution was requested.
5. **Damage functional constructed to produce result.** The $c_2/(V_t + \epsilon)$ term guarantees the conclusion by construction; independent justification needed.
6. **Endogeneity of friction.** Corruption can create friction, not just result from it; this feedback loop was unaddressed.
7. **Scope of $D_S$.** The automatable fraction of decision nodes was not estimated.
8. **Context-dependent moral costs.** The moral cost parameter $\mu_i$ should depend on institutional context, creating a feedback loop.
9. **Selection on the dependent variable.** Only confirming cases were cited.
10. **Time horizons and measurement.** Falsifiability criteria lacked specified time periods and measurement strategies.
11. **Minor issues.** Equation (6) assumed uniform friction distribution without stating it; $\kappa$ domain should be bounded; $\theta^*$ needed institutional interpretation; "proves" language too strong for assumption-dependent results; additional literature engagement needed.

*Revisions Made*

1. **Interaction structure.** Added Assumption 3.1 specifying repeated interaction, identifiability, and bounded network degree. Added explicit statement that the

theorem does not apply to one-shot anonymous interactions, with discussion of which institutional contexts satisfy the conditions.

2. **Classification uncertainty.** Added Remark 2.5 formalizing uncertainty in essential/discretionary classification through probabilistic rule assignment $q_j \in [0, 1]$. Derived expected friction and variance of friction estimates. Stated that downstream results hold in expectation but with confidence intervals determined by classification uncertainty.
3. $\Phi^*$ **estimation.** Added Remark 4.5 discussing indirect estimation strategies for $\Phi^*$ through bribe prices, time-cost comparisons, experimental moral cost elicitation, and observed corruption prevalence. Acknowledged that the theorem's practical force is primarily qualitative pending empirical calibration.
4. **Formal invariance bound.** Replaced "approximately invariant" with explicit bound (equation 24): $|\partial\gamma/\partial F| \leq g_{\max} \cdot p_d(\kappa^*(F))$, making volume sensitivity depend on the moral cost density at the participation threshold and the equilibrium detection probability. Stated conditions under which the bound is tight vs. loose.
5. **Damage functional justification.** Added Remark 7.4 providing three independent justifications for the $c_2/(V_t + \epsilon)$ form: Ashby's Law of Requisite Variety, historical evidence from institutionally rigid systems, and robustness to alternative functional forms. Explicitly acknowledged the result is conditional on this modeling choice.
6. **Endogeneity of friction.** Added new subsection in Limitations discussing the friction–corruption feedback loop, predicted bistability, and prioritization for future modeling.
7. **Scope of $D_S$.** Added new subsection in Limitations with crude estimates of automatable fractions and call for domain-specific empirical studies.
8. **Context-dependent moral costs.** Added new subsection in Limitations discussing endogenous moral cost dynamics and resulting path dependence.
9. **Selection on dependent variable.** Added new subsection in Limitations acknowledging the issue and calling for systematic empirical survey including challenging cases.
10. **Falsifiability time horizons.** Specified 2–5 year and 3–7 year time horizons for Principles I and II respectively. Added specific measurement strategies including proxy-based approaches for unobservable corruption volume.
11. **Minor issues.** Made uniform friction distribution assumption explicit in Theorem 3.4 proof. Replaced $\kappa \in [0, \infty)$ with $\kappa \in [0, \bar{\kappa}]$ in Definition 5.1 with discussion of bounded vs. unbounded concealment technology. Added interpretation of $\theta^*$

for institutional systems in Limitations. Added references to Fisman and Golden (2017), Chalfin and McCrary (2017), and Burgess et al. (2012).

### Subsequent Rounds

*Placeholder — to be completed as needed.*